Housing Markets and the Belief in Opportunity*

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April 29, 2026

Abstract

We study how housing-market signals shape beliefs about economic opportunity. Using a randomized information experiment in Singapore, we vary exposure to information about public housing prices and housing-related policies, then elicit beliefs about future mobility for children born at the bottom and top of the income distribution. Information about rising house prices and higher property taxes significantly lowers perceived upward mobility for bottom-origin children, while leaving beliefs about top-origin children largely unchanged. By contrast, information about slower price growth and expanded housing subsidies generates no corresponding increase in mobility beliefs. Our results point to reference-dependent belief formation: signals associated with worsening housing affordability reduce perceived opportunity, while partial improvements do little to restore optimism.

JEL Classification: D84, D63, R21, H31

Keywords: Social mobility expectations; housing markets; belief formation; reference dependence; policy signals

*This project was reviewed and approved in advance by the Institutional Review Board at Nanyang Technological University (IRB No.IRB-2025-548). The randomized controlled trial was registered with the AEA RCT Registry under ID AEARCTR-0016640. We gratefully acknowledge funding support from the Social Mobility Foundation Grant at the Lee Kuan Yew School of Public Policy, National University of Singapore.

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1 Introduction

Housing shapes access to economic opportunity across generations. It influences where families live, the schools and labor markets children encounter, and the assets households accumulate and pass on (Chetty et al. 2016; Chetty and Hendren 2018). Affordability is central to this role. When housing becomes more expensive relative to income, access to ownership, desirable neighborhoods, and asset accumulation may become harder for households without existing wealth. Rising house prices therefore have clear distributional implications: they increase wealth for existing owners, but may raise entry barriers for non-owners and make advancement appear more dependent on family resources, housing wealth, and prior access to asset ownership. This helps explain why governments devote substantial policy attention to housing affordability. Yet we know relatively little about whether households interpret housing-market developments and policy responses as signals about the prospects for upward mobility.

Understanding this belief channel is important because beliefs about mobility shape how individuals interpret inequality, evaluate public policy, and support redistribution (Benabou and Ok 2001; Alesina et al. 2018). Standard frameworks emphasize income dynamics, human capital accumulation, and returns to effort as determinants of mobility beliefs (Piketty 1995; Benabou 1996). We examine a complementary possibility: salient housing-market and policy information may shape perceived mobility by changing how individuals assess affordability barriers to advancement. From this perspective, housing prices and housing policies matter not only because they affect actual constraints, but also because they may shape perceptions of who can move up and who remains protected from falling behind.

In this study, we examine housing-market and policy signals as a widely observed source of information shaping beliefs about social mobility. Across advanced economies, housing costs have risen relative to incomes, making affordability a visible economic concern for many households. A growing literature shows that housing prices shape beliefs about future prices, wealth accumulation, and economic prospects (Kuchler et al. 2023; Armona et al. 2019). Governments have responded with a range of housing policies, including demand-side taxes, targeted subsidies, and programs to expand access to homeownership. It remains unclear how households interpret these developments when forming beliefs about economic opportunity. Do signals associated with rising affordability pressures reduce perceived upward mobility? Do signals pointing to more moderate affordability conditions increase optimism? And do these responses differ for beliefs about children born at the bottom versus the top of the income distribution?

To answer these questions, we conduct a randomized information experiment in Singapore. Our approach builds on a growing literature that uses survey experiments to study how individuals update beliefs in response to information about economic conditions and public policy (D'Acunto and Weber 2024; Haaland et al. 2023). Respondents are randomly exposed to information about public housing price dynamics and housing-related policies. The price treatments describe either rising resale prices or slower resale price growth. The policy treatments describe either expanded housing subsidies for first-time buyers or higher property taxes on owner-occupied residential properties. We then elicit beliefs about the future social mobility of children born at the bottom and top of the income distribution. This design allows us to estimate how housing-related information affects mobility beliefs while holding fixed respondents' own economic circumstances and prior experiences.

We find asymmetric belief updating. Information associated with rising affordability pressures, namely rising house prices and higher property taxes, lowers perceived upward mobility for bottom-origin children, while leaving beliefs about top-origin children largely unchanged. These treatments also affect broader economic expectations and reduce trust in government, suggesting that respondents may interpret housing-related pressures not only as changes in prices or taxes, but also as information about the government's ability to support access to economic opportunity. By contrast, information pointing to more moderate affordability conditions, including slower house price growth and expanded housing subsidies, generates more limited responses and does not significantly increase mobility beliefs. Overall, the results are consistent with reference-dependent belief formation: information that heightens affordability concerns receives greater weight than information suggesting that these concerns may be moderating.

Our experiment is conducted in Singapore, a setting well suited for studying how housing-market and policy signals are interpreted. Housing plays a central role in households' economic lives. More than 80 percent of residents live in public Housing & Development Board (HDB) flats, over 90 percent of these households own their homes, and public housing is the primary asset, savings vehicle, and channel of intergenerational wealth transmission for many citizens. Housing affordability has also become a focal concern, as HDB resale prices have risen sharply and often outpaced wage growth.[1] Crucially, Singapore combines extensive housing subsidies and access programs with housing-related taxes and purchase restrictions. This exposes households to policy signals that may either support affordability or heighten concerns about housing costs. Unlike settings that rely primarily on one type of intervention, Singapore uses these instruments simultaneously, allowing us to study how households interpret contrasting housing policy signals within a single institutional environment. This makes Singapore a natural setting for examin-

[1] HDB flats are publicly built housing units sold to eligible households on long-term leases, typically 99 years. After satisfying minimum occupancy requirements, owners may sell these units in a regulated resale market. Resale prices are market-determined transaction prices for existing HDB units and are distinct from subsidized prices of newly built flats sold directly by the government.

ing how housing-market developments and policy interventions shape beliefs about social mobility.

Formally, we implement a randomized information experiment that varies housing-related information to study how respondents form beliefs about social mobility. The experiment varies information along two dimensions that are central to housing affordability and widely observed by households: HDB resale price dynamics and housing-related government policies. This design allows us to examine whether respondents interpret different housing signals differently, and whether belief responses vary across the income distribution. Respondents are randomly assigned to one of five groups. The first two treatments provide information about housing market conditions. Treatment 1 (T1) informs respondents that HDB resale prices have risen, while Treatment 2 (T2) emphasizes that the growth rate of HDB resale prices has slowed in recent years. Two additional treatments provide information about housing policies. Treatment 3 (T3) describes an expansion of government housing subsidies for first-time homebuyers, while Treatment 4 (T4) reports an increase in property tax rates for owner-occupied residential properties. A fifth group serves as the control group and receives no housing-related information. Together, these treatments allow us to assess whether information associated with rising affordability pressures, such as rising prices and higher property taxes, is interpreted differently from information pointing to more moderate affordability conditions, such as slower price growth and expanded subsidies.

Our experiment shows that respondents update beliefs differently across housing-related information. Information associated with rising affordability pressures generates broader revisions across expectations. Information about rising house prices (T1) increases expected future house-price growth and expected inflation, while reducing trust in government. Information about higher property taxes (T4) also lowers trust in government, although its effects on macroeconomic expectations differ from those of the price-growth treatment. This common decline in trust suggests that respondents may interpret housing-related pressures not only as changes in prices or taxes, but also as information about the government's ability to support access to economic opportunity.

By contrast, information pointing to more moderate affordability conditions generates more limited responses. Information that house price growth has slowed (T2) lowers expectations about future house prices, but does not significantly affect beliefs about inflation, aggregate economic growth, or trust in government. Information about expanded housing subsidies (T3) produces no statistically significant changes in expectations about house prices, economic growth, inflation, or government trust. Taken together, these results are consistent with reference-dependent belief updating. Information associated with rising affordability pressures may push respondents further below their perceived affordability benchmark and therefore affect several belief dimensions. Information pointing to more moderate affordability conditions generates smaller revisions, suggesting that partial improvements may not be sufficient to shift beliefs back toward that benchmark. Overall, the pattern suggests that housing-related information is evaluated relative to prevailing affordability benchmarks, with information below those benchmarks receiving greater weight than information suggesting partial improvement.

These asymmetries in belief updating extend directly to perceptions of economic opportunity and social mobility. Information about rising house prices and higher property taxes lowers perceived upward mobility for bottom-origin children by about 0.10 rungs on a five-rung mobility ladder. In contrast, none of the treatments significantly affects beliefs about mobility for children born at the top of the income distribution. Information pointing to more moderate affordability conditions, including slower house price growth and expanded housing subsidies, also produces no statistically significant change in perceived mobility for either bottom- or top-origin children.

This pattern closely parallels the broader belief responses documented above. Information associated with rising affordability pressures reduces perceived prospects for advancement among those starting at the bottom, while information suggesting partial moderation in these pressures generates little corresponding revision. By contrast, perceived mobility at the top remains largely unchanged across all treatments. Overall, housing-related information shapes beliefs about advancement in an asymmetric and distribution-specific manner: affordability pressures matter most for perceived upward mobility from the bottom, while beliefs about persistence at the top are less responsive to these signals.

We interpret these patterns as consistent with reference-dependent expectation formation in beliefs about economic opportunity. Rising housing prices and higher property taxes appear to be interpreted as salient pressures that disproportionately depress beliefs about upward mobility at the bottom of the income distribution, while information suggesting more moderate affordability conditions does little to restore optimism. By contrast, beliefs about the prospects of those at the top appear more anchored in perceived structural advantage and are largely insensitive to housing-related information. Together, these findings suggest that housing-market and policy signals shape beliefs about opportunity in an asymmetric and distribution-specific manner.

To further pin down the mechanisms underlying belief updating, we complement the experimental design with open-ended survey questions that elicit respondents' reasoning in their own words (Haaland et al. 2025). Open-ended responses allow us to observe how individuals interpret housing price changes and housing-related policies as signals about opportunity and constraint, without imposing ex ante categories on their thought process. This approach helps distinguish between competing mechanisms, such as rational updating about affordability, reference-dependent interpretations of changing benchmarks, and salience-driven emphasis on losses versus gains.

Our findings are important for how housing markets and housing policies are interpreted within the broader study of economic opportunity, and for how governments assess the societal consequences of housing interventions. A large literature documents how housing prices and housing policies shape wealth accumulation, consumption, credit access, and residential sorting, through channels such as housing wealth effects, collateral constraints, and location choice (Sodini et al. 2023). A separate literature examines beliefs about social mobility, inequality, and economic opportunity, often using survey and experimental methods to study perceptions of advancement, fairness, and redistribution (Benabou and Ok 2001; Cruces et al. 2013). Despite the central role of housing in shaping economic outcomes, relatively little is known about whether and how housing market conditions and housing policies shape beliefs about social mobility itself, rather than realized outcomes.

This distinction is relevant for policy interpretation. Governments often justify housing policies, such as affordability measures, subsidies, or macroprudential regulations, in part by their expected effects on economic opportunity and perceived fairness. Yet existing research provides limited guidance on whether perceptions of opportunity respond symmetrically to housing-related gains and losses across the income distribution, or whether changes in housing affordability disproportionately affect beliefs about prospects at the bottom versus perceived advantages at the top. As a result, it remains unclear how housing affordability pressures and housing policy interventions translate into broader perceptions of inequality and economic opportunity, even when their effects on realized outcomes are well understood.

These issues also matter for how standard economic frameworks are applied in policy analysis. Canonical models of housing markets and social mobility emphasize affordability constraints, borrowing limits, and realized outcomes, while largely abstracting from how individuals form beliefs about mobility across the income distribution (Becker and Tomes 1979; Benabou 1996). In such frameworks, housing price changes and policy incentives primarily affect the constraints households face, with limited implications for perceptions of who advances or falls behind. Under this view, improvements and deteriorations in housing affordability would be expected to generate broadly similar belief responses regarding mobility at both the bottom and the top. The asymmetric belief responses documented in this paper suggest that policy evaluations based solely on realized outcomes may miss an important belief-based dimension of how housing markets shape perceived economic opportunity.

Related Literature. This paper contributes to several strands of literature. First, we relate to the literature on housing price expectations and their economic effects. A large body of work shows that beliefs about future house prices shape individual behavior, including homeownership decisions (Bailey et al. 2018; Ben-David et al. 2019; Bottan and Perez-Truglia 2025), mortgage choice (Bailey et al. 2019; De Stefani 2021), housing investment (Armona et al. 2019), and consumption (Lambertini et al. 2013; Qian 2023; Chopra et al. 2025). Using a randomized information experiment, Chopra et al. (2025) show that higher expected long-run house price growth has sharply asymmetric effects across tenure status, with renters reducing consumption substantially while homeowners' spending is largely inelastic. We extend this literature by showing that housing prices shape not only housing and consumption decisions, but also households' beliefs about long-run economic opportunity. Our contribution is to provide causal evidence that housing price movements affect expectations about social mobility, a forward-looking belief that lies outside the housing market itself.

Second, this paper relates to a growing literature on how salient economic information shapes household beliefs. A central insight from this work is that belief formation depends not only on realized economic outcomes, but also on how economic signals are framed, presented, and attended to (Coibion et al. 2022; D'Acunto et al. 2021). Agarwal et al. (2022) show that in Singapore, providing households with price information about higher-quality goods raises inflation expectations, illustrating how the content of information can shift beliefs even in the absence of changes in underlying fundamentals. Recent work applies these insights to housing and policy environments. Kuang et al. (2026) show that information about borrower-based macroprudential policies causally shifts housing market expectations and affordability perceptions, while Ahn et al. (2024) document that housing exposure shapes how households process monetary policy information. We build on this literature by treating housing prices as salient economic signals that influence beliefs about long-horizon outcomes, rather than as information about near-term prices or policy conditions alone.

Third, we contribute to the literature on beliefs about social mobility and fairness. A substantial body of work uses surveys and experimental methods to study perceptions of inequality, mobility, and support for redistribution (Cruces et al. 2013; Karadja et al. 2017; Alesina et al. 2018; Fisman et al. 2022; Moore et al. 2025). Karadja et al. (2017) show that relative income positions shape redistributive preferences, while Alesina et al. (2018) document systematic biases in beliefs about intergenerational mobility and show that more pessimistic mobility perceptions increase support for redistribution. A related strand emphasizes reference dependence in the formation of fairness and redistribution beliefs. Fisman et al. (2022) demonstrate that reference points shape redistributive behavior in controlled experiments, and Alesina et al. (2023) show that perceived relative income positions within salient reference groups influence fairness views and policy attitudes. We add to this literature by providing causal, market-based evidence from Singapore that housing market conditions and housing policies shape social mobility beliefs in a reference-dependent manner.

The remainder of the paper proceeds as follows. Section 2 outlines the experimental design and data. Section 3 presents the main results on how housing market signals shape expectations of social mobility and related beliefs. Section 4 examines the underlying mechanisms through a reference-dependent framework. Section 5 evaluates alternative explanations and discusses policy implications. Section 6 concludes.

2 Experimental Design and Data

This section describes the institutional context, experimental design, and data used to study how housing market developments and housing-related policies shape beliefs about social mobility and related expectations. First, we outline the institutional features of Singapore's housing market that render housing prices and housing policies salient signals of economic opportunity. We then describe the sampling strategy and survey implementation, the elicitation of baseline beliefs, and the randomized information treatments. Finally, we discuss the measurement of post-treatment beliefs, the construction of the main outcome variables, summary statistics for the analysis sample, and the integrity of the randomization.

2.1 Institutional Background: Housing Markets in Singapore

We begin with a brief overview of the institutional structure of Singapore's public housing market, which provides the context for the housing price and policy information used in the survey. Public housing in Singapore is administered by a centralized government agency, the Housing & Development Board (HDB), and operates within a tightly regulated framework that links housing access, government subsidies, and housing-related taxes. HDB flats account for roughly 80 percent of resident households and more than 90 percent of owner-occupied housing, making housing the dominant asset for most Singaporean households. Consequently, changes in housing prices and housing policy are highly visible and are widely interpreted as signals of economic opportunity and government support.

Within this institutional framework, access to newly built public housing is tightly regulated. Eligibility is restricted to citizen households that meet requirements related to family structure, income, and prior property ownership, and applicants typically face longer waiting times before newly built units become available. Moreover, households above specified income ceilings are ineligible to purchase new public housing altogether. New units are sold at government-determined subsidized prices and are subject to resale restrictions during a fixed minimum occupation period (Housing & Development Board 2023). Together, these features channel many households, including higher-income households and those with urgent housing needs, toward the secondary market, namely the public housing resale market.

Unlike the primary market, transactions in the public housing resale market occur at market-determined prices and under substantially broader eligibility conditions. Resale transactions involve previously occupied HDB units traded on a secondary market in which prices are negotiated between buyers and sellers rather than set by the government. Both Singapore Citizens and Permanent Residents are eligible to participate, and resale purchases are not subject to binding income ceilings. Prices in the resale market therefore reflect housing demand, credit conditions, and policy announcements, and serve as the primary price signal households observe when assessing housing affordability, wealth accumulation, and economic opportunity. For this reason, our analysis and experimental treatments focus on resale HDB prices and resale-related housing policies. The resale market is the main channel through which households experience housing price movements and respond to housing-related policy changes.

In addition to market prices, government policy directly shapes housing costs and access in the resale market through subsidies. First-time buyers in the resale market may qualify for substantial, means-tested government support through the Central Provident Fund (CPF), a mandatory individual savings system used to finance housing purchases. The Enhanced CPF Housing Grant provides income-linked subsidies that vary discretely with household earnings and can represent a large share of annual income (Central Provident Fund Board 2025). Because these subsidies are explicitly quantified and directly applied at the point of purchase, changes in subsidy policy are easily understood and widely discussed.

Housing policy also affects households through property taxation. Owner-occupied residential properties are subject to progressive property taxes based on assessed rental values, and recent reforms have increased marginal tax rates, particularly for higher-value properties. These taxes are billed annually, vary discretely with property values, and are frequently adjusted in conjunction with broader fiscal and redistribution debates. Consequently, housing-related taxes are interpreted not only as changes in housing costs, but also as signals of government priorities and redistribution.

Taken together, housing in Singapore functions as a central gateway to homeownership that is shaped directly by government policy and market prices. Eligibility thresholds, income-linked subsidies, progressive housing taxes, and resale prices play visible roles in determining access and costs, and generate clear, widely observed signals about economic opportunity. This institutional setting provides a clean context for isolating how housing price movements and housing-related policies shape beliefs about social mobility.

2.2 Experiment Set-Up

We field a single-wave online survey experiment to study how housing market information and policy framing shape beliefs about social mobility and related expectations in Singapore. The target population consists of adult Singapore residents, with eligibility restricted to Singapore Citizens or Permanent Residents aged 21 and above. Recruitment is conducted through Rakuten Insight, a large commercial survey platform widely used for academic and market research in Asia.

Rakuten Insight maintains an extensive panel of Singapore residents and supports quota-based sampling along key demographic dimensions. Respondents who opt into the study complete eligibility screens, and sampling quotas are imposed so that the final sample closely matches national population distributions by gender, age, income, marital status, race, and homeownership status. Respondents receive compensation only upon completing the full questionnaire, in the form of Rakuten reward points that can be redeemed for gifts or vouchers through the platform. The survey was conducted over a two-month window from October to November 2025 and recruited a total of 3,160 participants.

To ensure data quality and internal consistency, we apply a set of standard data-cleaning procedures. First, we exclude respondents with total interview times below 280 seconds, which likely reflect insufficient attention to survey instructions and information screens. Second, we drop observations with internally inconsistent housing reports, specifically cases in which respondents indicate living in owner-occupied housing while reporting ownership of no properties. Third, we trim extreme outliers in reported past housing price beliefs by excluding respondents who report annual price changes of 60 percent or more. Finally, we exclude respondents in the policy treatment arms who misinterpret the direction of the policy signal. Together, these restrictions reduce noise arising from inattention, misunderstanding, and implausible inputs[2]. Online Appendix reports the complete survey questionnaire.

The survey follows a three-stage structure. Respondents first report baseline beliefs prior to any information exposure. They are then randomly assigned either to one of the information treatments, which provide housing price or housing policy signals, or to a pure control group that receives no information. Finally, respondents report post-treatment beliefs, which constitute the main outcomes of interest.

[2]We conduct a robustness check using a more conservative sample construction to verify that our main results are not driven by sample construction. Specifically, we apply only two sample restrictions. First, we drop respondents who provide internally inconsistent housing tenure/ownership reports. Second, we exclude interviews with very short completion times.

2.2.1 Elicitation of Prior Beliefs

Before any information is shown, we elicit a comprehensive set of baseline expectations. These priors capture beliefs about social mobility at both the individual and societal levels, as well as the broader macroeconomic and housing context in which mobility assessments are formed. Specifically, we measure (i) perceived mobility prospects for one's own household, (ii) beliefs about intergenerational mobility for bottom and top origin groups, and (iii) expectations about housing prices, macroeconomic conditions, and government performance that may co-move with mobility beliefs.

Social mobility beliefs are elicited using a set of social position ladder tasks following Alesina et al. (2018). Figure A.1 in the Online Appendix provides an illustration of these tasks. For perceived individual upward mobility, the ladder consists of ten rungs, where rung 1 represents households that are worst off in Singapore and rung 10 represents those that are best off. Respondents first place their own household on the ladder based on its current position and then indicate where they expect their household to be ten years in the future. This task provides a simple and intuitive measure of perceived individual upward mobility over time.

Intergenerational mobility is captured through a related allocation task. Respondents are shown a five-rung social ladder and asked to consider 100 children born today to families in the bottom twenty percent of the current income distribution. They allocate these children across the five rungs according to where they expect them to be as adults, with allocations required to sum to 100. The same exercise is then repeated for 100 children born to families in the top twenty percent of the income distribution. The resulting allocations generate full destination distributions that summarize expected upward mobility from the bottom of the income distribution and persistence or downward mobility from the top. These measures constitute the primary outcomes in our analysis.

Beyond mobility beliefs, we collect baseline expectations that characterize the broader economic and housing environment. Respondents report perceived changes in HDB resale prices in recent years and state expectations for house price changes over the next twelve months using a probability-bin elicitation that sums to 100. We additionally collect one-year-ahead expectations for aggregate economic growth and inflation, and measure baseline trust in the Singapore government to do what is right to enhance social mobility on a 1-5 scale.

2.2.2 Treatment

Following the elicitation of baseline beliefs, respondents are exposed to a randomized information intervention. Respondents are assigned either to receive factual information or to a pure control group that receives no information, allowing us to identify the causal effects of housing market signals and housing-related policy information on beliefs while holding fixed baseline views. The intervention introduces exogenous variation along two dimensions that are central to housing affordability and widely observed by households: housing market conditions and housing-related government policies. Respondents are randomly assigned to one of four information treatments or to a baseline control group.

The four treatments are organized into two conceptually parallel pairs. As discussed earlier, we focus on the resale public housing market, which dominates Singapore's housing system and renders price movements and housing-related policies salient signals of economic opportunity. The first pair varies information about recent housing price dynamics, capturing market-generated signals about access to housing and asset accumulation. The second pair varies information about housing-related government policies, capturing policy-generated signals about government support for, or constraints on, access to housing. Across all treatments, information is drawn from official sources, presented using factual descriptions and statistics.

Housing Price Treatments. We begin with two treatments that provide information about recent housing price developments. Treatment 1 (T1) highlights a sustained increase in housing prices in recent years, while Treatment 2 (T2) emphasizes a deceleration in housing price growth. Figure 1 displays the housing price information shown to respondents in each treatment.

<Insert Figure 1 Here>

In T1, respondents see Panel (a) in Figure 1, which shows that the HDB Resale Price Index has increased markedly over time. Rising housing prices constitute a salient market signal of deteriorating affordability and tighter access to a key asset associated with economic advancement. In the context of social mobility, such price increases may be interpreted as raising barriers to entry and delaying or foreclosing pathways to asset accumulation. This treatment therefore allows us to study how adverse housing market signals shape beliefs about upward social mobility, particularly for individuals starting at the bottom of the income distribution, and whether beliefs about mobility at the top respond in a similar manner.

In Treatment 2 (T2), respondents receive the information summarized in Panel (b) of Figure 1. A deceleration in housing price growth may signal a moderation in housing market pressures. At the same time, when price levels remain elevated, slower growth need not translate into a meaningful expansion of access to housing. This treatment therefore allows us to examine whether more favorable or neutral housing market signals generate corresponding revisions in beliefs about upward social mobility, or whether beliefs respond asymmetrically to positive and negative housing developments.

Housing Policy Treatments. The remaining two treatments shift attention from market outcomes to government interventions in the housing market. Treatment 3 (T3) highlights an expansion of government support for first-time homebuyers. In contrast, Treatment 4 (T4) emphasizes an increase in housing-related taxes. Figure 2 displays the information shown to respondents in these treatments.

<Insert Figure 2 Here>

In T3, respondents receive information about an expansion of housing subsidies for first-time buyers. Specifically, respondents are informed that the maximum Enhanced CPF Housing Grant has increased from **S$80,000** to **S$120,000**, and that total grants available to eligible first-time buyers can now reach up to **S$230,000**. This information is presented in tabular form, as shown in Figure 2 Panel (a).

Here, expanded housing subsidies represent a policy signal intended to ease access to homeownership and mitigate affordability constraints. This treatment allows us to assess whether information about policy support is interpreted as expanding economic opportunity and improving prospects for upward mobility, or whether such policy signals are perceived as insufficient to alter underlying beliefs about mobility.

In T4, respondents receive information about an increase in housing-related taxes. Respondents are informed that property tax rates for owner-occupied residential properties have risen from 2024 onward. For homes with an annual value of **S$100,000**, the annual property tax payable has increased from **S$8,730** to **S$11,980**. In addition, the highest marginal tax rate on owner-occupied residential properties has increased from **23 percent** to **32 percent**. This information is presented in a format parallel to Treatment 3, as shown in Figure 2 Panel (b).

Higher housing-related taxes constitute a policy signal that increases the cost of holding housing assets and may be interpreted as tightening constraints on wealth accumulation. This treatment allows us to examine whether policy-induced increases in housing costs shape beliefs about upward social mobility in a manner comparable to adverse housing market developments, and whether beliefs about mobility at different points of the income distribution respond symmetrically to market and policy signals.

2.2.3 Post-Treatment Beliefs

After the information is shown, we elicit respondents' post-treatment expectations using a format that closely parallels the baseline. In order to reduce consistency pressures and mitigate mechanical anchoring to earlier answers, we vary the phrasing and response scale of several questions while keeping the underlying constructs comparable. Expectations about housing prices, inflation, and macroeconomic conditions are collected as point forecasts rather than probability distributions. Perceptions of social mobility are measured using the same ladder framework as in the baseline, but respondents are now asked to allocate outcomes for 300 children rather than 100 and to reassess their own household's position ten years ahead. This allows us to capture genuine belief updating rather than repetition of previously stated values.

2.3 Data

We now turn to our data. Table 1 reports summary statistics for the main variables used in the analysis. The analysis sample consists of 2,299 respondents who completed the survey and passed standard data cleaning procedures. The table summarizes demographic characteristics, socioeconomic status, and subjective expectations elicited before and after information exposure.

<Insert Table 1 Here>

The sample spans a wide age range, with a mean age of approximately 45 years, and includes substantial variation in household size, income, and educational attainment. Gender and marital status are balanced, and a large majority of respondents report being employed. Housing characteristics indicate that most respondents live in public housing, reflecting the dominant role of HDB housing in Singapore, while a smaller share reside in private housing. The table also reports respondents' birthplace and parental education, capturing background characteristics relevant for beliefs about intergenerational mobility. Exact variable definitions and coding are reported in the table notes. This table further documents respondents' baseline expectations prior to receiving any information. On average, respondents expect housing prices to increase by about 4.4 percent over the next year. Baseline expectations for aggregate economic growth average around 2.0 percent, while expected inflation averages 3.9 percent. Trust in government with respect to enhancing social mobility is measured on a five-point scale and has a mean of 3.44.

Social Mobility Data. We summarize respondents' intergenerational mobility beliefs using an expected-rung measure. For each respondent i, we compute the expected rung as

$$ER_i = \sum_{r=1}^{5} r \cdot P_{ir} \tag{1}$$

where P_{ir} is the proportion of children that respondent i assigns to rung r (the proportions sum to 1). A higher ER_i means a higher expected social position. We compute this measure separately for children born to families in the bottom 20 percent and for children born to families in the top 20 percent of the social distribution. For bottom-origin children, higher expected rung values indicate greater perceived upward mobility. For top-origin children, higher expected rung values indicate stronger expected persistence at the top of the distribution.

Prior to information exposure, respondents place their own household at an average of 6.13 on the ten-rung social ladder. For intergenerational mobility, the expected rung for children born to families in the bottom 20 percent of the social distribution averages 2.86 on the five-rung ladder, while the expected rung for children born to families in the top 20 percent averages 3.71. The table further reports the same expected-rung measures after information exposure.

Integrity of Randomization. We assess the integrity of the randomization by testing for balance in observable characteristics across treatment and control groups. Appendix Table A.1 reports differences in means between each treatment group and the control group for a broad set of demographic, socioeconomic, and pre-treatment belief variables. Baseline social mobility beliefs are balanced across treatment and control groups. In particular, prior expected rung for bottom-origin children shows no statistically significant differences across groups, with p-values ranging from 0.105 to 0.908. Expected rungs for top-origin children are similarly balanced, with p-values between 0.173 and 0.720, as are respondents' expectations about their own future social position, for which p-values range from 0.288 to 0.900.[3]

3 Main Results

3.1 Determinants

We begin with a descriptive analysis of the cross-sectional relationships between mobility beliefs, macroeconomic expectations, and individual characteristics, to provide context for the experimental results that follow. Table A.2 in the Online Appendix reports pairwise correlations among mobility beliefs, macroeconomic expectations, and demographic characteristics. Housing price expectations exhibit a pronounced asymmetry across mobility dimensions. Higher expected housing prices are negatively correlated with perceived individual mobility (-0.175, $p < 0.01$) and bottom-origin mobility (-0.053, $p < 0.05$), but positively correlated with top-origin mobility (0.235, $p < 0.01$). This pattern indicates that rising housing prices are associated with perceived advantages for those already at

[3]Figures A.2 and A.3 present tests of balance for baseline beliefs using cumulative distribution functions (CDFs).

the top, while simultaneously weakening perceived opportunities for others, motivating the focus on housing price signals in the experimental design.

Furthermore, macroeconomic expectations are meaningfully related to mobility beliefs. Expected economic growth is positively correlated with higher mobility, while expected inflation is negatively correlated with perceived individual mobility (-0.176, $p < 0.01$) and bottom-origin mobility (-0.064, $p < 0.01$). Government trust is strongly positively correlated with perceived individual upward mobility (0.297, $p < 0.01$) and bottom-origin mobility (0.179, $p < 0.01$), and negatively correlated with top-origin mobility (-0.088, $p < 0.01$).

This evidence suggests that beliefs about social mobility are closely intertwined with broader perceptions of economic conditions and institutional effectiveness, highlighting the importance of accounting for these channels when examining how housing-related information shapes mobility expectations. This section presents the main empirical results from the survey experiment. We begin with visual evidence on pre- and post-treatment belief updating across information arms. We then document the correlates of mobility beliefs and estimate the causal effects of housing market and policy signals on economic expectations. Finally, we examine treatment effects on intergenerational mobility beliefs and present evidence on the perceived channels linking housing conditions to social mobility.

3.2 Pre- vs Post-Treatment Beliefs: Visual Evidence.

We present visual evidence on the main empirical patterns documented in the data. Figures 3-5 plot the distributions of pre-treatment and post-treatment beliefs across the four treatment arms for respondents' expected mobility for their own household, expected upward mobility for bottom-origin children, and expected persistence for top-origin children. Each panel overlays the pre- and post-treatment histograms for the corresponding treatment arm.

<Insert Figure 3 Here>

The visual evidence indicates that belief updating is concentrated among expectations for bottom-origin mobility in the adverse signal treatments. In particular, Panels A and D of Figure 4 show a leftward shift in post-treatment beliefs following information about rising housing prices (T1) and higher housing-related taxes (T4). In contrast, the corresponding distributions in the cooling housing price growth (T2) and subsidy (T3) treatments exhibit only weak or negligible changes, suggesting limited belief updating in response to these more neutral or supportive signals. For beliefs about one's own future mobility and about persistence among top-origin children, the pre- and post-treatment distributions remain largely stable across all treatment arms. While small shifts can be detected in some panels, these movements are modest relative to the dispersion of beliefs and do not display a consistent directional pattern across treatments. This visual stability suggests that information shocks primarily affect perceptions of vulnerability and upward mobility at the bottom of the distribution, rather than expectations about one's own position or the durability of advantage at the top.

<Insert Figure 4 Here>

Across all treatment arms, expected mobility for bottom-origin children is consistently evaluated at substantially lower levels than expected persistence for top-origin children, as seen by comparing Figures 4 and 5. This ordering aligns with respondents' baseline perceptions of intergenerational inequality and indicates that the treatments operate on a shared and coherent belief structure rather than generating implausible or mechanically induced shifts. The clear separation between bottom- and top-origin distributions thus supports the internal integrity of the experimental design and the validity of the elicited belief measures.

<Insert Figure 5 Here>

3.3 Housing Signals and Economic Expectations

Next, we analyze treatment effects on respondents' expectations about the housing market and the macroeconomic environment. For this analysis, we focus on post-information expectations of housing price growth over the next 12 months, aggregate economic growth, perceived inflation, and trust in government. We estimate separate OLS regressions for each outcome, comparing each treatment group to the control group while controlling for demographic characteristics and baseline expectations. Specifically, we estimate treatment effects using the following specification:

$$Y_i^{\text{post}} = \alpha + \beta\,\text{Treat}_i + \gamma' X_i + \varepsilon_i, \tag{2}$$

where Y_i^{post} denotes respondent i's post-information expectation. Depending on the specification, Y_i^{post} corresponds to expected housing price growth over the next 12 months, expected aggregate economic growth, expected inflation, or trust in government. Treat_i is an indicator equal to one if respondent i is assigned to the relevant information treatment and zero if assigned to the control group. X_i is a vector of control variables including demographic characteristics and baseline expectations for the corresponding outcome. The coefficient β captures the causal effect of the information treatment on post-information expectations.

<Insert Table 2 Here>

Table 2 reports the estimated treatment effects on post-information expectations. We begin by examining responses in expected housing price growth, reported in Columns (1)–(4). Exposure to information emphasizing rising housing prices (T1) increases expected price growth by 0.884 percentage points, statistically significant at the 1% level. Conversely, information highlighting slower housing price growth (T2) leads to a reduction in expected price growth of 1.456 percentage points, also statistically significant at the 1% level. By contrast, information about housing subsidies (T3) and property taxes (T4) does not generate statistically significant changes in housing price expectations, indicating that respondents sharply differentiate between market-based price signals and policy interventions when forming price expectations.

We next turn to expectations about broader macroeconomic growth, reported in Columns (5)–(8). Consistent with the price-expectation results, respondents exposed to rising housing price information (T1) revise expected economic growth upward by 0.585 percentage points. None of the remaining treatments lead to statistically significant changes in expected growth, with point estimates close to zero and relatively precisely estimated. This pattern suggests that housing price signals, but not housing policy signals, translate into revisions in perceived aggregate economic performance.

We then examine inflation expectations in Columns (9)–(12). Information about rising housing prices (T1) increases expected inflation by 0.399 percentage points, consistent with respondents interpreting price increases as reflecting broader inflationary pressures. In contrast, information about higher property taxes (T4) leads to a modest decline in expected inflation of 0.340 percentage points, statistically significant at the 10% level. Information about slower price growth and housing subsidies does not significantly affect inflation expectations. Together, these results indicate that price- and policy-based housing signals can move inflation expectations in opposite directions.

Finally, we assess whether the treatments also affect institutional confidence. Columns (13)–(16) report treatment effects on trust in government. Exposure to rising housing price information reduces trust in government, with the effect statistically significant at the 10% level. Information about higher property taxes generates a larger and statistically significant decline in trust. No statistically significant effects are observed for the remaining treatments. Notably, declines in trust arise under both adverse price and adverse policy treatments, despite their differing implications for macroeconomic expectations.

Overall, the experimental treatments produce strongly asymmetric belief responses. Information about adverse housing developments leads to sizable and coordinated revisions across multiple belief dimensions, whereas favorable signals generate little adjustment. When respondents are informed that house prices have risen (T1), they revise expectations of future house prices and inflation upward and report a significant decline in trust in government. Exposure to higher housing taxes (T4) produces a comparable decline in trust, despite respondents revising expected inflation and economic growth downward. Thus, reductions in trust arise under both adverse price and policy treatments, even though these signals imply opposite movements in macroeconomic expectations.

3.4 Treatment Effects on Intergenerational Mobility

In what follows, we examine how the information treatments affect beliefs about intergenerational social mobility for children born to families at different points of the social distribution. As described above, we summarize respondents' mobility beliefs using the expected-rung measure constructed from the five-rung social mobility ladder. We compute expected rung separately for children born to families in the bottom 20 percent and in the top 20 percent of the social distribution. Our main outcome of interest is the post-information expected rung. We estimate treatment effects using the following specification:

$$ER_{ig}^{\text{post}} = \alpha + \beta\,\text{Treat}_i + \gamma' X_i + \varepsilon_{ig}, \tag{3}$$

where ER_{ig}^{post} denotes respondent i's post-information expected rung for children from origin group $g \in \{\text{bottom}, \text{top}\}$. Treat_i is an indicator equal to one if respondent i is assigned to the relevant information treatment and zero if assigned to the control group. X_i is the same vector of covariates as in the previous specification. The coefficient β captures the causal effect of the information treatment on intergenerational mobility beliefs.

<Insert Table 3 Here>

Table 3 reports treatment effects on post-treatment beliefs about intergenerational mobility[4]. We first examine mobility beliefs for children born to families at the bottom in order to assess whether housing-related information is interpreted as constraining upward mobility. Columns (1)-(4) show mobility expectations for bottom-origin children. Exposure to information about rising housing prices (T1) leads to a statistically significant decline in expected mobility for children born to families in the bottom 20 percent, with the expected rung falling by 0.101 relative to the control group (statistically significant at the 5% level). Similarly, exposure to information about higher property taxes (T4) produces a negative revision in bottom-origin mobility beliefs. Specifically, respondents revise the expected rung attained by children born to families in the bottom 20%

[4]Table A.3 reports main estimation coefficients from the alternative sampling. The restricted sample yields coefficients of similar magnitude that remain statistically significant.

downward by approximately 0.104 rungs relative to the control group, statistically significant at the 5% level. In contrast, information about cooling housing price growth (T2) and housing subsidies (T3) does not produce statistically significant changes in expected bottom-origin mobility, with point estimates close to zero.

We then turn to mobility beliefs for children born to families at the top of the income distribution in order to assess whether the same housing-related information alters perceptions of persistence at the top. Columns (5)-(8) present results for top-origin children. Across all four treatments, the estimated effects on expected top-origin mobility are small in magnitude and statistically insignificant. Information about housing prices, housing policies, or property taxes does not lead respondents to revise beliefs about mobility persistence at the top of the distribution.

Taken together, the results reveal a clear asymmetry in belief updating across origin groups. Housing-related information systematically affects perceptions of mobility for children starting at the bottom of the income distribution, while leaving beliefs about mobility at the top largely unchanged. For bottom-origin children, responses are uniformly pessimistic when information highlights either rising housing costs or increased fiscal burdens associated with housing. Overall, these patterns are consistent with reference-dependent belief formation, in which adverse housing signals disproportionately shape perceptions of mobility constraints at the bottom, with little corresponding revision in expectations at the top.

3.5 Treatment Effects on Perceived Individual Upward Mobility Beliefs

We also examine whether the information treatments affect respondents' perceptions of their own future social mobility. In particular, we compare each treatment group to the pure control group. The outcome of interest is respondents' post-information assessment of their own future social position, measured on the ten-rung social ladder.

We estimate the following regression specification:

$$Y_i^{\text{post}} = \alpha + \beta \, \text{Treat}_i + \gamma' X_i + \varepsilon_i, \tag{4}$$

where Y_i^{post} denotes respondent i's post-information perceived future social position, Treat_i is an indicator equal to one if respondent i is assigned to the relevant information treatment and zero if assigned to the control group, and X_i is a vector of covariates including demographic characteristics and baseline personal mobility beliefs. The coefficient β captures the causal effect of the information treatment on personal mobility beliefs.

Table A.4 in the Online Appendix reports treatment effects on respondents' perceptions of their own future social mobility. Across all four information treatments, the estimated effects are small in magnitude and statistically insignificant. Information about rising housing prices (T1), cooling price growth (T2), housing subsidies (T3), and property taxes (T4) does not lead to revisions in respondents' self-assessed future social position.

This pattern is consistent with a reference-dependent framework. Respondents differ in their baseline social position, housing status, and life stage, implying that the same housing market or policy signal is evaluated against different benchmarks. As a consequence, identical information can imply positive deviations from the reference point for some individuals and negative deviations for others. When belief updating is reference-dependent in this way, these opposing revisions can offset each other in the cross-section, leading to small and statistically insignificant average treatment effects. This mechanism explains why information that meaningfully shifts beliefs about aggregate or intergenerational mobility does not translate into detectable average changes in perceived individual upward mobility.

3.6 Perceived Channels Linking Housing and Social Mobility

To provide additional evidence on how individuals conceptualize the link between housing and social mobility, Figure 6 summarizes respondents' assessments of factors they considered when forming views about people's social standing or upward mobility. Respondents evaluated five housing-related channels: home values, living costs, borrowing costs, government action, and overall economic conditions, rating each on a five-point scale from "not at all" to "extremely."

<Insert Figure 6 Here>

The figure reveals clear heterogeneity in the perceived importance of different channels. Living costs, borrowing costs, and overall economic conditions receive the largest shares of responses in the "very" and "extremely" categories. This pattern indicates that respondents primarily associate upward mobility with affordability, access to credit, and broader economic conditions that shape households' ability to accumulate resources. These channels are closely linked to day-to-day financial constraints and are therefore viewed as central to moving up the social ladder.

By contrast, home values (housing prices) and government action (housing policy) attract substantially fewer top-end responses. While these channels are not viewed as irrelevant, respondents are considerably less likely to regard rising housing prices or government intervention as key drivers of upward mobility. In other words, improvements in housing prices or policy conditions are not widely perceived as primary mechanisms through which individuals advance economically.

Importantly, because the survey question explicitly focuses on upward mobility, the figure is silent about whether these same channels matter for downward mobility or perceived risks of falling behind. In particular, the weaker association between housing prices and upward mobility does not imply that housing market conditions are unimportant. Rather, it suggests that respondents do not view favorable housing price movements as generating large mobility gains. Our experimental evidence directly complements this pattern by showing that housing price signals appear to operate primarily on the downside, increasing perceived risks of reduced opportunity rather than enhancing expectations of upward mobility. Taken together, the survey and experimental results point to an asymmetric role of housing market signals: they are weakly associated with beliefs about upward mobility, but strongly shape perceptions of downside risk. We build on this insight in the next section.

4 Conceptual Framework: Reference Dependence

This section develops a reference-dependent framework to explain the asymmetric belief updating across housing signals and origin groups. We demonstrate that the heterogeneity in responses across socioeconomic characteristics and prior beliefs is consistent with reference-dependent belief formation.

4.1 Interpreting Asymmetric Belief Updating: Reference Dependence

Our empirical results show that belief updating in response to these signals is strongly asymmetric. Adverse housing signals, such as rising housing costs or higher housing-related taxes, generate large and coordinated revisions in beliefs, while favorable housing signals generate little response. This asymmetry is difficult to reconcile with interpretations based solely on affordability constraints, borrowing limits, or wealth effects. Under such views, improvements and deteriorations in housing conditions should lead to broadly symmetric belief revisions, and beliefs at the bottom and the top of the income distribution should move in parallel. Neither pattern is observed in the data.

A natural way to interpret these findings is through reference-dependent belief formation. Households evaluate housing market developments relative to prevailing benchmarks shaped by recent conditions and lived experience. These benchmarks reflect commonly held expectations about affordability, access to ownership, and the feasibility of economic advancement. New information is interpreted relative to these reference points rather than processed symmetrically around a fixed standard.

Within this framework, adverse housing signals are interpreted as losses relative to existing benchmarks. Such signals are especially salient when they reinforce the perception that economic advancement increasingly depends on asset ownership that is out of reach for many households. As a result, negative housing signals exert a strong influence on beliefs about upward mobility, particularly at the bottom of the income distribution. By contrast, favorable housing signals, such as housing subsidies or slowing price growth, generate weaker belief responses when they are perceived as insufficient to materially relax constraints relative to prevailing affordability challenges. Improvements that do not restore affordability to prior benchmarks do little to reverse pessimism about long-run opportunity.

The same logic explains why beliefs about mobility at the top of the income distribution are largely insensitive to housing-related signals. For higher-income households, access to housing is less constrained and housing affordability is less informative about long-run opportunity. Perceived advantage is instead anchored in factors such as education, networks, and occupational status, which are only weakly affected by housing market conditions. As a result, housing-related signals carry limited informational content for beliefs about mobility at the top.

Overall, housing market developments and housing-related policies shape beliefs about economic advancement in a strongly asymmetric and distribution-specific manner. Adverse signals substantially depress perceived upward mobility at the bottom, favorable signals fail to offset pessimism, and beliefs about mobility at the top remain largely unchanged.

4.2 Reference Dependent Model

To formalize the asymmetric responses documented above, we adopt a reference-dependent formulation following Kőszegi and Rabin (2006). Housing-market information is modeled as shifting perceived mobility prospects relative to benchmark distributions formed by prior beliefs. We interpret reported mobility beliefs as evaluative beliefs about attainable mobility rather than as statistical forecasts. In this interpretation, respondents map perceived mobility prospects into reported beliefs through a reference-dependent evaluation function. Shifts in $U_i(\delta)$ therefore correspond to shifts in perceived opportunity.

Setup Let $x \in \mathbb{R}$ denote a perceived future mobility outcome and $r \in \mathbb{R}$ denote a reference outcome. Instantaneous reference-dependent utility is

$$u(x \mid r) = m(x) + \mu\big(m(x) - m(r)\big), \tag{5}$$

where $m(\cdot)$ is increasing and differentiable. The gain-loss component is

$$\mu(y) = \begin{cases} \eta y, & y \geq 0, \\ \eta\lambda y, & y < 0, \end{cases} \qquad \eta > 0,\ \lambda > 1, \tag{6}$$

so that outcomes below the reference point receive greater marginal weight. Individuals evaluate uncertain mobility prospects by comparing an outcome lottery F to a reference lottery G:

$$U(F \mid G) = \iint u(x \mid r)\, dG(r)\, dF(x). \tag{7}$$

A housing-market treatment delivers a signal $s \in \mathbb{R}$. The signal shifts respondent i's perceived mobility prospects, represented as a change from outcome lottery F_{i0} to F_{is}. To accommodate heterogeneity in benchmarks, let G_{i0} denote respondent i's pre-signal reference lottery. Over the short experimental horizon, reference points are locally fixed, so the relevant object is

$$U_i(F_{is} \mid G_{i0}). \tag{8}$$

To obtain transparent comparative statics, we impose a location-shift structure. Let $X_{i0} \sim F_{i0}$ and assume

$$X_{is} = X_{i0} + \delta_i(s), \qquad X_{is} \sim F_{is}, \qquad \delta_i(0) = 0, \tag{9}$$

with $\delta_i'(s) > 0$. Define

$$U_i(\delta) \equiv U_i(F_{i\delta} \mid G_{i0}). \tag{10}$$

Boundary Sensitivity Heterogeneity arises from differences in how perceived outcomes relate to individual reference benchmarks. Let $R_{i0} \sim G_{i0}$ denote a draw from respondent i's reference lottery and define the comparison gap

$$Z_i \equiv m(X_{i0}) - m(R_{i0}), \tag{11}$$

which measures perceived distance from the benchmark. Under a signal-induced shift,

$$Z_{i,\delta} \equiv m(X_{i0} + \delta) - m(R_{i0}). \tag{12}$$

We specialize to the linear case

$$m(x) = x, \tag{13}$$

so that $Z_i = X_{i0} - R_{i0}$ and $Z_{i,\delta} = Z_i + \delta$. Let $H_i(\cdot)$ and $h_i(\cdot)$ denote the CDF and density of Z_i. Define respondent i's boundary sensitivity as

$$\varphi_i \equiv h_i(0), \tag{14}$$

the density of comparison gaps near the gain-loss boundary. The level $H_i(0)$ captures the share of perceived outcomes already in the loss region, while the density $h_i(0)$ captures how much probability mass lies near the gain-loss boundary. Thus, immediate marginal sensitivity depends on $H_i(0)$, whereas $h_i(0)$ determines how quickly this sensitivity changes around the reference point. Higher φ_i implies that small adverse shifts reclassify more realizations from gains to losses. Hence, the reference-dependent utility can be written as

$$U_i(\delta) = \mathbb{E}[X_{i0}] + \delta + \mathbb{E}\big[\mu(Z_i + \delta)\big]. \tag{15}$$

Proposition 1 (Asymmetric Response to Housing Signals)**.** *Suppose the comparison gap Z_i has continuous CDF $H_i(\cdot)$. Then the marginal effect of an outcome shift δ on respondent i's reference-dependent utility is*

$$\frac{dU_i(\delta)}{d\delta} = 1 + \eta + \eta(\lambda - 1)H_i(-\delta). \tag{16}$$

For adverse signals ($\delta < 0$), $H_i(-\delta)$ increases relative to favorable shifts, so the amplification term $\eta(\lambda - 1)H_i(-\delta)$ becomes larger. This generates stronger downward revisions in perceived mobility when substantial probability mass lies below or near the reference benchmark. For favorable signals ($\delta > 0$), $H_i(-\delta)$ is smaller and marginal utility collapses toward $1 + \eta$.

Proposition 1 shows that favorable signals generate weaker marginal responses than adverse signals. If favorable housing signals are small or perceived as insufficient to move outcomes above the reference benchmark, the resulting belief revision may be empirically muted.

Proposition 2 (Boundary Sensitivity and Bottom-Origin Vulnerability)**.** *The curvature of reference-dependent utility is*

$$\frac{d^2U_i(\delta)}{d\delta^2} = -\eta(\lambda - 1)\, h_i(-\delta), \tag{17}$$

and at the benchmark $\delta = 0$,

$$\left.\frac{d^2 U_i(\delta)}{d\delta^2}\right|_{\delta=0} = -\eta(\lambda - 1)\varphi_i. \tag{18}$$

The strength of this asymmetry depends on the density of comparison gaps around the gain-loss boundary. Individuals whose perceived mobility prospects are close to their reference benchmark have high $h_i(0)$. For them, small adverse housing signals move more probability mass into the loss region, increasing the decline in perceived opportunity. This provides a natural interpretation for why bottom-origin mobility beliefs are especially responsive to adverse housing signals.

In the housing context, this logic suggests that individuals who view homeownership as a just-attainable benchmark are especially sensitive to adverse housing signals. Their perceived outcomes are concentrated near the gain-loss boundary, implying a high $\phi_i = h_i(0)$. For a small adverse shift $\delta < 0$, the share of realizations reclassified from gains to losses is approximately $\phi_i(-\delta)$, leading to a disproportionately large decline in perceived opportunity relative to individuals whose outcomes lie farther from the boundary.

The following corollary characterizes how differences in reference benchmark distributions G_{i0} translate into differences in $H_i(-\delta)$ and therefore into heterogeneous belief responses to the same housing signal. Consider two individuals i and j who receive the same housing-market signal s and share identical gain-loss preferences (η, λ). Suppose heterogeneity arises through differences in reference benchmarks G_{i0}.

Corollary 1 (Heterogeneous Reference Benchmarks)**.** *If individual i holds a more demanding reference benchmark than individual j, meaning R_{i0} is higher so that $Z_i = X_{i0} - R_{i0}$ tends to take lower values, formally $H_i(z) \geq H_j(z)$ for all z, then*

$$\frac{dU_i(\delta)}{d\delta} \geq \frac{dU_j(\delta)}{d\delta}.$$

Since adverse signals correspond to a negative change in δ, a larger marginal derivative implies a larger decline in utility or perceived opportunity for the same adverse shift.

This suggests that people who believe structural barriers such as family backgrounds and connections are primary determinants of mobility set a higher bar for what constitutes an improvement, placing more of their perceived outcomes in the loss region and amplifying the utility decline under adverse housing signals. This demanding benchmark implies that favorable signals are insufficient to push perceived outcomes above the reference point, accounting for the absence of upward belief revision in this group.

4.3 Heterogeneity in Belief Updating

To further validate the reference-dependent interpretation, this subsection examines heterogeneity in belief updating across socioeconomic characteristics and prior beliefs. The boundary sensitivity and reference benchmarks provide a unifying framework for interpreting the heterogeneous treatment effects documented in this section. Fragility captures the extent to which perceived mobility outcomes are concentrated near the gain-loss boundary, making beliefs particularly sensitive to adverse housing signals. We show that identical housing signals generate sharply different responses depending on respondents' income, liquidity, education, and perceived social position, consistent with belief updating relative to group-specific reference points.

Intergenerational Heterogeneity: Education. Table 4 examines heterogeneity in belief updating by educational attainment, separately for beliefs about upward mobility from the bottom and persistence at the top. High education is defined as the completion of at least post-secondary education.

For beliefs about bottom-origin mobility, average treatment effects are small and statistically insignificant, but there is meaningful heterogeneity by educational attainment. Across columns (1)-(4), interaction terms with high education are consistently negative and modest in magnitude, indicating more pessimistic updating among highly educated respondents. In particular, for the rising housing price treatment (T1), the interaction with high education is -0.216 rungs and statistically significant at the 5% level, implying that highly educated respondents revise beliefs about bottom-origin mobility downward relative to less educated respondents. A similar pattern emerges for the tax treatment (T4), where the interaction term is -0.167 rungs and statistically significant at the 10% level. Taken together, these results suggest that highly educated respondents respond more pessimistically to adverse housing signals.

On the other hand, education heterogeneity is strong and statistically significant for beliefs about top-origin mobility. In columns (5)-(7), the baseline treatment effects are negative, indicating that respondents with lower education revise beliefs about top persistence downward following exposure to housing information. In contrast, the interaction terms associated with high education are positive and statistically significant at the 5% level for T1 and T2 and at the 1% level for T3, with coefficients between 0.221 and 0.326 rungs. This implies that highly educated respondents revise beliefs about top-origin mobility upward relative to less educated respondents when exposed to the same information.

<Insert Table 4 Here>

This pattern indicates that educational attainment shapes the reference point against which housing signals are interpreted: less educated respondents evaluate signals relative to beliefs that top advantage is contestable, while highly educated respondents reference beliefs about structural persistence. Highly educated respondents evaluate housing market and policy information relative to a reference point that emphasizes institutional stability and the persistence of advantage. From this perspective, rising prices, cooling growth, and subsidies are interpreted as reinforcing the mechanisms through which advantage at the top is maintained, such as asset ownership, policy design, and market structure, rather than as forces that equalize opportunities. Consequently, these signals increase expected persistence at the top without improving expectations about mobility from the bottom. By contrast, respondents with lower education are more likely to interpret the same information relative to a reference point centered on potential redistribution or equalization. Housing market interventions and policy actions may therefore be viewed as limiting elite advantage or increasing fluidity, leading to downward revisions in beliefs about top persistence.

Intergenerational Heterogeneity: Family and Connections. Next, we turn to heterogeneity by respondents' beliefs about the role of family background and social connections in shaping social mobility. Table 5 and 6 examine heterogeneity in mobility belief updating by respondents' beliefs about the importance of family background and social connections in shaping social mobility, defined as those ranking these factors 4 and 5 in importance. This classification captures a subjective assessment of how opportunity is structured, distinguishing respondents who view mobility as strongly shaped by inherited advantage from those who place less weight on such factors.

<Insert Table 5 Here>

For bottom-origin mobility, baseline treatment effects are small and statistically insignificant across all four treatments. However, the interaction terms with family background are consistently negative and statistically significant. Specifically, the interaction coefficients are -0.185 rungs for T1 (statistically significant at the 5% level), -0.176 rungs for T2 (statistically significant at the 1% level), and -0.145 rungs for T3 (statistically significant at the 5% level), with a smaller and statistically insignificant effect for T4. A similar pattern emerges for beliefs about social connections: the interaction terms are negative and statistically significant for the subsidy and tax treatments, with magnitudes of -0.172 rungs for T3 and -0.164 rungs for T4. These results indicate that respondents who believe family background or connections matter revise expectations about upward mobility from the bottom downward relative to other respondents when exposed to housing-related information.

<Insert Table 6 Here>

For top-origin mobility, the pattern is symmetric but opposite in sign. Baseline treatment effects are negative across all treatments, indicating that respondents who do not emphasize family background or connections revise beliefs about persistence at the top downward. In contrast, the interaction terms with family background are positive and statistically significant for all four treatments, with coefficients of 0.184 rungs for T1 (statistically significant at the 5% level), 0.178 rungs for T2 (statistically significant at the 5% level), 0.253 rungs for T3 (statistically significant at the 1% level), and 0.239 rungs for T4 (statistically significant at the 1% level). Interaction effects for social connections are similarly positive and statistically significant for the subsidy and tax treatments, with magnitudes of 0.197 and 0.204 rungs, respectively. These estimates indicate that respondents who view family background or connections as important revise expectations about top persistence upward relative to others.

Respondents who believe mobility is strongly determined by family background or connections evaluate housing-related information relative to a reference point in which advantage is inherited and institutionally reinforced. Relative to this benchmark, housing price movements and policy interventions are interpreted not as equalizing forces, but as signals that existing hierarchies will persist. As a result, information exposure reinforces pessimism about opportunities at the bottom and optimism about persistence at the top. By contrast, respondents who place less weight on family background or connections evaluate the same information relative to a more meritocratic reference point, leading to weaker or opposite revisions. While baseline specifications yield statistically insignificant average treatment effects on perceived individual upward mobility, the heterogeneity results reveal large and systematic differences in belief updating across income, liquidity and self-evaluation groups.

Individual Upward Mobility Heterogeneity: Income and Liquidity We next focus on perceived individual upward mobility and examine whether treatment responses differ by income. The main treatment coefficient captures the effect among lower-income respondents, while the interaction term captures the differential response among higher-income respondents.

<Insert Table 7 Here>

For low-income respondents, all four treatments reduce perceived individual upward mobility. Three of the four effects are statistically significant and economically meaningful. Information about cooling housing price growth (T2) lowers expected individual mobility by 0.221 rungs, information about housing subsidies (T3) reduces expected mobility by 0.349 rungs, and information about higher property taxes (T4) reduces expected mobility by 0.32 rungs. The estimate for rising housing prices (T1) is also negative but insignificant. In contrast, high-income respondents exhibit systematically attenuated or offsetting responses. The interaction terms are positive and statistically significant at the 1% level for T2, T3, and T4, with coefficients of 0.430, 0.585, and 0.413 rungs, respectively. While low-income respondents revise perceived individual mobility beliefs downward, these statistically significant interaction effects indicate that high-income respondents revise their perceived individual mobility beliefs upward in response to the same information.

Table A.5 in the Online Appendix examines heterogeneity by housing liquidity which proxies for the ability to cover three months of living costs if an unexpected expense arose. Respondents without liquidity represent those for whom housing market and policy shocks are more likely to bind, while liquid respondents possess sufficient buffers to absorb such shocks. For illiquid respondents, all four treatments reduce perceived individual mobility, with large and statistically significant effects in most cases. Rising housing prices (T1) reduce perceived individual mobility by 0.566 rungs, statistically significant at the 1% level. Cooling price growth (T2) lowers expected mobility by 0.419 rungs, statistically significant at the 10% level. The largest effect arises for the subsidy treatment (T3), which reduces perceived individual mobility by 0.847 rungs. Higher property taxes (T4) also reduce perceived mobility by 0.408 rungs. In contrast, liquid respondents exhibit sharply different responses. The interaction terms are positive and statistically significant at the 1% level for T1 and T3, and at the 5% level for T2, with coefficients of 0.665, 0.949, and 0.477 rungs, respectively. These results indicate that respondents with sufficient liquidity revise perceived individual mobility beliefs upward relative to illiquid respondents when exposed to the same information. For the tax treatment (T4), the interaction term is positive but not statistically significant, suggesting that fiscal extraction signals are less strongly differentiated by liquidity.

For low-income respondents, the reference point is centered on the feasibility of upward mobility through housing, specifically, whether homeownership and asset accumulation constitute a realistic path to advancement. Relative to this benchmark, all four treatments are interpreted as failing to restore meaningful opportunity. Information about rising housing prices (T1) signals worsening affordability, but also carries ambiguous implications about future asset values, which weakens salience and yields an insignificant estimated effect. Cooling price growth (T2) does not relax binding constraints, and therefore represents stagnation rather than improvement, leading to a modest but negative belief revision. The strongest responses occur for policy-based treatments. Subsidy information (T3) is interpreted not as a gain, but as evidence that housing affordability has become so strained that access increasingly depends on policy discretion; when perceived as insufficient or uncertain, subsidies highlight the persistence of structural barriers and generate downward revisions in perceived individual mobility beliefs. Higher property taxes (T4) similarly signal government extraction and tighter conditions for asset accumulation, reinforcing perceptions of constrained advancement.

For high-income respondents, the reference point is fundamentally different. It is defined not by the feasibility of entry, but by asset preservation, financial flexibility, and control over timing. Relative to this benchmark, price signals (T1 and T2) carry limited negative implications. Rising prices imply little erosion of purchasing power and may preserve relative advantage, while cooling price growth reduces volatility and downside risk, increasing the option value of waiting and stabilizing future prospects. Policy interventions (T3 and T4) are likewise interpreted as positive, either because benefits are additive rather than binding, or because fiscal changes pose limited threat to advancement. Consequently, high-income respondents revise individual mobility beliefs upward relative to low-income respondents, as reflected in large and highly statistically significant interaction effects.

Individual Upward Mobility Heterogeneity: Bottom Perception. Table 8 examines heterogeneity in individual upward mobility belief updating by respondents' perceived position in the social distribution, distinguishing those who perceive themselves to be near the bottom from those who do not. This dimension captures a subjective reference point that is distinct from income or liquidity but closely related to perceived vulnerability and exposure to downside risk.

For respondents not perceiving themselves as bottom, all four treatments are associated with positive and statistically significant revisions in perceived individual upward mobility. The estimated treatment effects range from 0.488 to 0.738 rungs and are statistically significant at the 1% level across T1-T4. This indicates that, for respondents who view themselves as relatively secure, housing market and policy information is interpreted as an opportunity for personal advancement. In sharp contrast, respondents who perceive themselves as bottom exhibit large and statistically significant downward revisions in individual upward mobility beliefs across all treatments. The interaction terms between bottom perception and treatment are uniformly negative and statistically significant at the 1% level. The interaction effects are economically large and statistically significant, ranging from -1.438 to -1.876 rungs across treatments.

<Insert Table 8 Here>

These patterns are strongly consistent with reference-dependent belief updating. Respondents who perceive themselves as near the bottom evaluate housing-related information relative to a reference point defined by acute vulnerability and limited scope for upward mobility. Relative to this benchmark, all four treatments, whether price-based or policy-based, signal that constraints remain binding or that advancement depends on forces beyond individual control. As a result, the information is coded as a loss, producing sharp downward revisions in perceived individual mobility.

By contrast, respondents who do not perceive themselves as bottom evaluate the same information relative to a reference point characterized by security and control. Relative to this benchmark, rising prices, cooling growth, subsidies, and even higher taxes carry limited negative implications and may signal stability or policy engagement, resulting in upward belief revisions.

5 Discussion

In this section, we show that alternative explanations such as wealth effects, affordability considerations, and generalized economic sentiment cannot account for the observed patterns. We then discuss the implications of belief-based responses for the design and communication of housing policies.

5.1 Alternative Explanations

We now evaluate three alternative channels through which housing-related information could affect beliefs about social mobility: wealth effects, affordability-based channels, and general economic pessimism or optimism. These channels differ in how housing market information is mapped into beliefs about opportunity and therefore imply distinct empirical predictions. While each channel captures certain features of the data, none can account for the pronounced asymmetry in belief updating across signals, outcomes, and socioeconomic groups documented above.

Wealth Effects. One alternative interpretation is that the observed belief updating reflects a pure wealth effect, whereby information about housing prices or housing-related policies mechanically alters perceived household wealth and thereby shifts expectations about social mobility. Under this view, rising housing prices or housing subsidies should increase perceived mobility by raising expected asset values, while cooling price growth or higher housing-related taxes should reduce perceived mobility by lowering expected wealth. This mechanism predicts that belief updating should be strongest among households with greater housing exposure or asset holdings, and that responses should be broadly symmetric for favorable and adverse signals.

Several features of the results are inconsistent with this interpretation. First, belief updating is concentrated in perceptions of upward mobility for bottom-origin children, rather than in beliefs about persistence at the top of the income distribution, where wealth-based effects should be most salient. Second, high-income and high-liquidity respondents, who would experience smaller expected capital losses when housing prices cool, revise mobility beliefs upward in response to cooling price growth information. In contrast, low-income respondents revise mobility beliefs downward even in response to information about housing subsidies. These patterns are difficult to reconcile with a mechanism driven primarily by changes in perceived household wealth.

Affordability-Based Channels. A second explanation is that housing information affects beliefs about social mobility through perceived changes in affordability and living costs. Under this mechanism, rising housing prices or higher housing-related taxes tighten affordability constraints and should reduce perceived upward mobility, while cooling price growth or housing subsidies relax constraints and should increase perceived mobility. Unlike wealth effects, affordability-based channels operate through current and expected budget constraints and therefore predict belief updating that is symmetric around improvements and deteriorations in affordability.

While affordability considerations are clearly salient for respondents, the empirical patterns are not fully consistent with this channel alone. Information about cooling housing price growth and housing subsidies does not systematically increase perceived upward mobility for bottom-origin children, despite representing clear improvements in affordability. More broadly, a pure affordability-based model implies symmetric belief updating around changes in constraints. If housing-related information primarily operates by relaxing or tightening budget constraints, then improvements and deteriorations of comparable magnitude should generate correspondingly symmetric revisions in perceived mobility. The sharp asymmetry observed in the data, where adverse housing signals consistently depress perceived mobility while favorable signals generate little response, is difficult to reconcile with such a framework. These patterns indicate that housing information is not interpreted solely through immediate affordability constraints, but also through broader assessments of opportunity structure and persistence.

General Economic Pessimism or Optimism. A third possibility is that housing-related information shifts respondents' overall economic sentiment, which then spills over into beliefs about social mobility. Under this view, treatments that convey adverse housing conditions would induce broad pessimism, leading to uniformly lower mobility beliefs across outcomes and groups, while more favorable information would generate generalized optimism.

The evidence does not support this interpretation. First, belief updating is not uniform across outcomes. Responses are substantially stronger for intergenerational mobility, particularly for bottom-origin children, than for beliefs about one's own future mobility, which would be equally affected under a general sentiment channel. Second, respondents exposed to the same information often revise beliefs in opposite directions depending on income, liquidity, education, and beliefs about the structural determinants of mobility. Such sharply heterogeneous responses are difficult to reconcile with a common shift in mood or sentiment.

Taken together, these findings indicate that housing-related information does not operate primarily by changing perceived household wealth, symmetrically relaxing or tightening affordability constraints, or shifting overall economic sentiment. Instead, the evidence points to belief formation that depends on how housing market signals are interpreted relative to salient reference points. Adverse housing signals receive disproportionate weight in shaping perceptions of opportunity, particularly for those at the bottom of the income distribution, while corresponding improvements are discounted. This interpretation accounts for both the asymmetry in average belief updating and the pronounced heterogeneity across socioeconomic and belief groups, and highlights the importance of accounting for belief formation when assessing the broader social impacts of housing market developments and policy interventions.

5.2 Policy Implications

Our results suggest that housing market conditions and housing-related policies shape beliefs about social mobility primarily through how households interpret signals about economic opportunity, rather than through mechanical wealth or affordability effects alone. Adverse housing developments and policy changes that are perceived as tightening access or increasing housing-related costs disproportionately depress perceived upward mobility at the bottom of the income distribution, while comparable improvements generate little offsetting optimism. Consequently, policies intended to expand access or reduce housing costs may not translate into commensurate improvements in perceived opportunity, particularly among groups that view advancement as structurally constrained.

This asymmetry has important implications for the design and evaluation of housing policies. Governments often justify housing interventions, such as affordability measures, subsidies, or property taxation, in part by their expected effects on economic opportunity and fairness. The evidence here suggests that policy communication and framing may be as important as policy content itself. When housing policies are experienced as salient losses or as signals of tighter constraints, they may undermine perceptions of opportunity even when they are intended to improve affordability. Conversely, policies that generate gradual or less visible gains may fail to shift beliefs if they do not reset salient reference points or alter how households perceive access to homeownership.

Importantly, these belief responses are conceptually distinct from the real effects of housing policies on prices, quantities, or household balance sheets. Even when policies succeed in easing affordability constraints or expanding access in equilibrium, households may continue to perceive opportunity as limited if policy interventions are interpreted as evidence of persistent structural barriers. In this sense, belief responses matter independently of realized outcomes. Perceptions of mobility shape political attitudes, trust in institutions, and support for redistribution, and may influence long-run behavior through channels such as human capital investment, savings decisions, and residential choice. Evaluations that focus exclusively on realized affordability or ownership outcomes may therefore understate the broader social consequences of housing market developments and policy interventions.

These findings align with models of belief formation that emphasize reference dependence and salience. In such frameworks, negative deviations from salient benchmarks receive disproportionate weight, while comparable gains are discounted. Rising housing prices and housing-related taxes thus function as salient losses relative to reference points anchored in affordability and expected access to homeownership, particularly for households at the bottom of the income distribution. By contrast, improvements in housing conditions may be less salient or insufficient to reset these reference points, yielding little upward revision in perceived mobility. From this perspective, housing markets influence beliefs about economic opportunity not through symmetric updating about fundamentals, but through asymmetric responses to perceived losses versus gains.

Finally, it is important to clarify the scope and external validity of these findings. Singapore is not intended to be representative of housing markets more broadly. Its centralized housing system, near-universal homeownership, and strong government presence render housing prices and housing policies unusually salient and widely interpreted as signals of economic opportunity. These institutional features allow us to isolate belief formation mechanisms in a setting where housing-related information is highly visible and policy-relevant, rather than to assert that the magnitude of the estimated effects will necessarily generalize to other contexts.

At the same time, the mechanisms highlighted in this paper are not inherently specific to Singapore. Reference dependence, loss asymmetry, and salience in belief formation are broad features of how individuals process economic information. In many advanced economies, housing prices, property taxation, and affordability policies are among the most prominent and widely discussed economic signals, even in more decentralized housing systems. To the extent that households elsewhere interpret housing market developments and policy interventions as signals about opportunity and constraint, similar asymmetric belief responses may arise. What is likely to be context-specific is the strength and clarity of these signals, which depend on institutional features such as policy credibility, market transparency, and the role of housing in household balance sheets. Singapore therefore serves as a clean laboratory for identifying these mechanisms, rather than a special case to which they are confined.

6 Conclusion

In sum, this paper provides direct evidence on how housing market signals and housing-related policies shape beliefs about social mobility. Using a survey experiment that exogenously varies information about house price changes, housing taxes, and housing subsidies, we elicit expectations about future mobility for individuals at the bottom and at the top of the income distribution. Three findings emerge. First, increases in house prices and housing-related taxes significantly reduce perceived upward mobility at the bottom of the income distribution, while leaving beliefs about mobility at the top largely unchanged. Second, symmetric favorable signals, such as slower house price growth and housing subsidies, do not raise perceived mobility at the bottom. Third, beliefs about mobility at the top appear largely insulated from both housing market developments and housing policy interventions. Together, these results show that housing affordability pressures shape perceptions of economic opportunity in an asymmetric and distribution-specific manner, operating primarily through beliefs about constraints at the bottom rather than perceived advantages at the top.

More broadly, the findings highlight a belief-based channel through which housing markets influence perceptions of opportunity and fairness, beyond their direct effects on realized outcomes such as wealth accumulation or access to homeownership. Standard economic frameworks typically evaluate housing policies through their impacts on constraints, incentives, and distributional outcomes. The evidence here suggests that housing prices and housing-related policies also shape how households interpret the scope for advancement across the income distribution. When adverse housing signals are salient and widely observed, they disproportionately depress optimism about upward mobility, while comparable improvements fail to restore confidence. Accounting for this belief-based channel may therefore be important for understanding the full societal consequences of housing market developments and housing policy interventions.

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Tables and Figures

Table 1: Summary Statistics

Variable	N	Mean	SD	Min	Max
Panel A: Demographics					
Age	2299	45.403	12.648	21	81
Household Size	2299	2.415	1.424	0	16
Female	2299	0.496	0.500	0	1
Marital Status	2299	0.679	0.467	0	1
Income	2299	8937.582	6138.055	0	22000
Education	2299	0.829	0.377	0	1
Father Education	2299	0.419	0.493	0	1
Mother Education	2299	0.294	0.456	0	1
Employed	2299	0.873	0.333	0	1
Housing Type	2299	0.216	0.412	0	1
Birthplace	2299	0.828	0.377	0	1
Panel B: Prior Expectations					
Housing Price Expectations (Pre)	2299	4.354	3.666	-10	10
Economic Growth Expectations (Pre)	2299	2.018	3.195	-10	10
Inflation Expectations (Pre)	2299	3.873	3.305	-10	10
Trust Expectation (Pre)	2299	3.440	0.896	1	5
Bottom Mobility (Pre)	2299	2.858	0.726	1	5
Top Mobility (Pre)	2299	3.713	0.781	1	5
Individual Mobility (Pre)	2299	6.130	1.853	1	10

Notes: This table reports summary statistics for demographics and prior expectations. The sample includes 2,299 respondents. Demographic variables include age, household size, gender, marital status, household income, education, parental educations, employment status, housing type, and birthplace. Expectation variables include prior expectations about housing prices, economic growth, inflation, trust for government, social mobility for top and bottom 20% children, and participants' perceived individual upward mobility. We recode survey responses into a set of indicator and continuous variables used as controls. Female is an indicator equal to 1 if the respondent reports being female and 0 if male. Marital Status equals 1 if the respondent reports being married and 0 otherwise. Income is elicited in categories and recoded as a continuous midpoint measure. Specifically, unemployed/no income is coded as 0; "Below 2,000" as 1,000; 2,000-3,999 as 3,000; 4,000-4,999 as 4,500; 5,000-6,999 as 6,000; 7,000-8,999 as 8,000; 9,000-10,999 as 10,000; 11,000-12,999 as 12,000; 13,000-14,999 as 14,000; 15,000-19,999 as 17,500; and 20,000 and above as 22,000. Education is coded into an indicator equal to 1 for respondents who report higher educational attainment (Polytechnic diploma or above) and 0 otherwise (primary, secondary, or junior college). Parental education is constructed analogously using father's and mother's highest education. Employed is an indicator equal to 1 for respondents who are employed (full-time, part-time, or self-employed) and 0 otherwise (unemployed, student, or not in the labor force). Housing type equals 1 if the respondent lives in private housing and 0 otherwise. Finally, Birthplace is an indicator equal to 1 if the respondent reports being born in Singapore and 0 otherwise. Prior expectations are constructed from baseline belief measures. Housing price, economic growth, and inflation expectations are constructed as weighted averages from probability-bin responses. Government trust is measured on a 1–5 scale and used as reported. Mobility beliefs are summarized as expected-rung measures from ladder allocations. Individual mobility refers to respondents' expected own social position ten years in the future. Top Mobility and Bottom Mobility refer to expected-rung measures for children born to families in the top and bottom 20% of the social position distribution, respectively.

Table 2: Treatment Effects on Macro Expectations

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)	(14)	(15)	(16)
	Housing Price Expectation				Economic Growth Expectation				Inflation Expectation				Trust Expectation			
T1	0.884***				0.585***				0.399**				-0.088*			
	(0.208)				(0.212)				(0.201)				(0.048)			
T2		-1.457***				-0.262				-0.279				0.063		
		(0.228)				(0.206)				(0.199)				(0.047)		
T3			0.250				0.156				0.186				0.037	
			(0.212)				(0.210)				(0.201)				(0.045)	
T4				-0.140				-0.337				-0.340*				-0.130***
				(0.214)				(0.216)				(0.206)				(0.049)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	917	935	939	891	917	935	939	891	917	935	939	891	917	935	939	891
R^2	0.107	0.106	0.091	0.106	0.265	0.233	0.279	0.267	0.172	0.153	0.153	0.130	0.401	0.352	0.379	0.392

Notes: This table reports OLS estimates of the effects of housing-related information treatments on post-information macroeconomic expectations, including housing price growth, economic growth, inflation, and government trust. Columns (1)-(4) examine expectations about housing price growth, columns (5)-(8) examine expectations about economic growth, columns (9)-(12) examine expectations about inflation, and columns (13)-(16) examine expectations about government trust. Each column corresponds to a separate regression comparing one treatment group to the control group. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; and T4 provides information about property taxes. All specifications include the same set of controls including age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, race, and pre-treatment expectations of the macro variables. Robust standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 3: Treatment Effects on Intergenerational Social Mobility Expectations

	Bottom-Origin Mobility				**Top-Origin Mobility**			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1	-0.101**				0.007			
	(0.049)				(0.051)			
T2		-0.015				-0.009		
		(0.048)				(0.050)		
T3			-0.032				-0.027	
			(0.047)				(0.049)	
T4				-0.104**				-0.002
				(0.049)				(0.051)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	917	935	939	891	917	935	939	891
R^2	0.091	0.103	0.105	0.106	0.041	0.055	0.041	0.039

Notes: This table reports OLS estimates of the effects of housing-related information treatments on intergenerational mobility expectation for children born to families at the bottom and top 20% of the distribution. Columns (1)–(4) use mobility expectations for bottom-origin children as the dependent variable, while columns (5)–(8) use mobility expectations for top-origin children. Each column corresponds to a separate regression comparing one treatment group to the control group. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; and T4 provides information about property taxes. All specifications include the same set of controls including age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, race, and pre-treatment expectations about economic growth conditions and trust. Robust standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table 4: Treatment Effects on Intergenerational Social Mobility Expectations: Heterogeneity by Education Level

	Bottom-Origin Mobility				**Top-Origin Mobility**			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1	0.080				-0.182*			
	(0.083)				(0.103)			
High Education × T1	-0.216**				0.226**			
	(0.088)				(0.107)			
T2		0.099				-0.193**		
		(0.085)				(0.090)		
High Education × T2		-0.137				0.221**		
		(0.088)				(0.093)		
T3			0.052				-0.301***	
			(0.088)				(0.104)	
High Education × T3			-0.099				0.326***	
			(0.091)				(0.107)	
T4				0.034				-0.056
				(0.095)				(0.100)
High Education × T4				-0.167*				0.061
				(0.098)				(0.104)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	917	935	939	891	917	935	939	891
R^2	0.093	0.102	0.104	0.107	0.033	0.047	0.031	0.033

Notes: This table reports OLS estimates of the heterogeneous effects of housing-related information treatments on intergenerational social mobility expectations by education. Columns (1)–(4) use mobility expectations for bottom-origin children as the dependent variable, while columns (5)–(8) use mobility expectations for top-origin children. Each column corresponds to a separate regression comparing one treatment group to the control group. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; and T4 provides information about property taxes. High education is defined as completing at least post-secondary education (Polytechnic diploma or above). All specifications include the same set of controls, including age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, race, and pre-treatment expectations about economic growth and government trust. Robust standard errors are reported in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table 5: Treatment Effects on Intergenerational Social Mobility Expectations: Heterogeneity by Family Wealth Beliefs

	Bottom-Origin Mobility				Top-Origin Mobility			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1	0.025				-0.119			
	(0.070)				(0.073)			
Family Wealth × T1	-0.185**				0.184**			
	(0.075)				(0.079)			
T2		0.097				-0.123*		
		(0.062)				(0.068)		
Family Wealth × T2		-0.176***				0.178**		
		(0.067)				(0.072)		
T3			0.064				-0.195***	
			(0.064)				(0.067)	
Family Wealth × T3			-0.145**				0.253***	
			(0.067)				(0.072)	
T4				-0.044				-0.162**
				(0.067)				(0.070)
Family Wealth × T4				-0.090				0.239***
				(0.072)				(0.075)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	917	935	939	891	917	935	939	891
R^2	0.097	0.109	0.109	0.107	0.046	0.061	0.053	0.049

Notes: This table reports OLS estimates of the heterogeneous effects of housing-related information treatments on intergenerational mobility expectations by beliefs about family wealth advantage. Columns (1)–(4) use mobility expectations for bottom-origin children as the dependent variable, while columns (5)–(8) use mobility expectations for top-origin children. Each column corresponds to a separate regression comparing one treatment group to the control group. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; and T4 provides information about property taxes. Family Wealth is an indicator equal to one if the respondent rates family wealth& background as 4 or 5 on a five-point importance scale. All specifications include the same set of controls, including age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, race, and pre-treatment expectations about economic growth and government trust. Robust standard errors are reported in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 6: Treatment Effects on Intergenerational Social Mobility Expectations: Heterogeneity by Connection Beliefs

	Bottom-Origin Mobility				**Top-Origin Mobility**			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1	-0.062				-0.005			
	(0.071)				(0.073)			
Connection × T1	-0.055				0.016			
	(0.075)				(0.079)			
T2		0.067				-0.031		
		(0.068)				(0.072)		
Connection× T2		-0.117				0.031		
		(0.072)				(0.076)		
T3			0.090				-0.167**	
			(0.067)				(0.070)	
Connection × T3			-0.172**				0.197***	
			(0.070)				(0.074)	
T4				0.008				-0.140*
				(0.067)				(0.072)
Connection × T4				-0.164**				0.204***
				(0.073)				(0.077)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	917	935	939	891	917	935	939	891
R^2	0.091	0.105	0.110	0.110	0.041	0.055	0.048	0.046

Notes: This table reports OLS estimates of the heterogeneous effects of housing-related information treatments on intergenerational mobility expectations by beliefs about social connections. Columns (1)–(4) use mobility expectations for bottom-origin children as the dependent variable, while columns (5)–(8) use mobility expectations for top-origin children. Each column corresponds to a separate regression comparing one treatment group to the control group. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; and T4 provides information about property taxes. Social connections are indicators equal to one if the respondent rates connections & networks as 4 or 5 on a five-point importance scale. All specifications include the same set of controls, including age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, race, and pre-treatment expectations about economic growth and government trust. Robust standard errors are reported in parentheses. *, **, *** denote significance at the 10%, 5%, and 1% levels.

Table 7: Treatment Effects on Perceived Individual Upward Mobility: Heterogeneity by Income

	Perceived Individual Upward Mobility			
	(1)	(2)	(3)	(4)
T1	-0.111			
	(0.125)			
High Income × T1	0.176			
	(0.152)			
T2		-0.221*		
		(0.126)		
High Income × T2		0.430***		
		(0.146)		
T3			-0.349***	
			(0.135)	
High Income × T3			0.585***	
			(0.163)	
T4				-0.316**
				(0.127)
High Income × T4				0.413***
				(0.155)
Controls	Yes	Yes	Yes	Yes
N	917	935	939	891
R^2	0.229	0.217	0.212	0.286

Notes: This table reports OLS estimates of heterogeneous treatment effects on perceived individual upward mobility by income. The dependent variable is respondents' expected own social position ten years in the future. Each column corresponds to a separate regression comparing one treatment group to the control group. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; and T4 provides information about property taxes. High income is an indicator equal to one for respondents with monthly household income of SGD 9,000 or above. All specifications include the same set of controls, including age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, race, and pre-treatment expectations about economic growth and government trust. Robust standard errors are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table 8: Treatment Effects on Perceived Individual Upward Mobility Expectations: Heterogeneity by Perceived Bottom Status

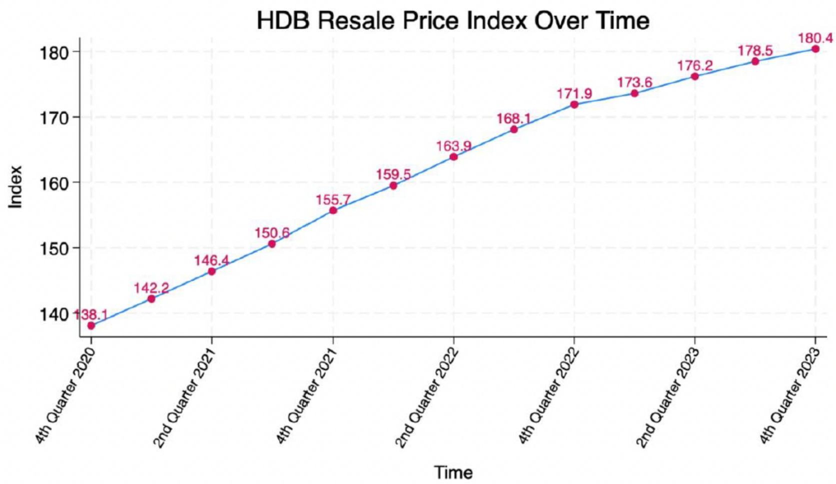
	Perceived Individual Upward Mobility			
	(1)	(2)	(3)	(4)
T1	0.578***			
	(0.105)			
Bottom Status × T1	-1.451***			
	(0.138)			
T2		0.592***		
		(0.100)		
Bottom Status × T2		-1.480***		
		(0.137)		
T3			0.738***	
			(0.100)	
Bottom Status × T3			-1.876***	
			(0.145)	
T4				0.488***
				(0.103)
Bottom Status× T4				-1.438***
				(0.143)
Controls	Yes	Yes	Yes	Yes
N	917	935	939	891
R^2	0.331	0.327	0.358	0.372

Notes: This table reports OLS estimates of heterogeneous treatment effects on perceived individual upward mobility by self-perceived bottom status. The dependent variable is respondents' expected own social position ten years in the future. Each column corresponds to a separate regression comparing one treatment group to the control group. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; and T4 provides information about property taxes. Self-perceived bottom status is an indicator equal to one if the respondent places their current position in the bottom 2 rungs of the social ladder. All specifications include the same set of controls, including age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, race, and pre-treatment expectations about economic growth and government trust. Robust standard errors are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Figure 1: Housing Price Information Treatments

(a) T1: Rising Housing Price

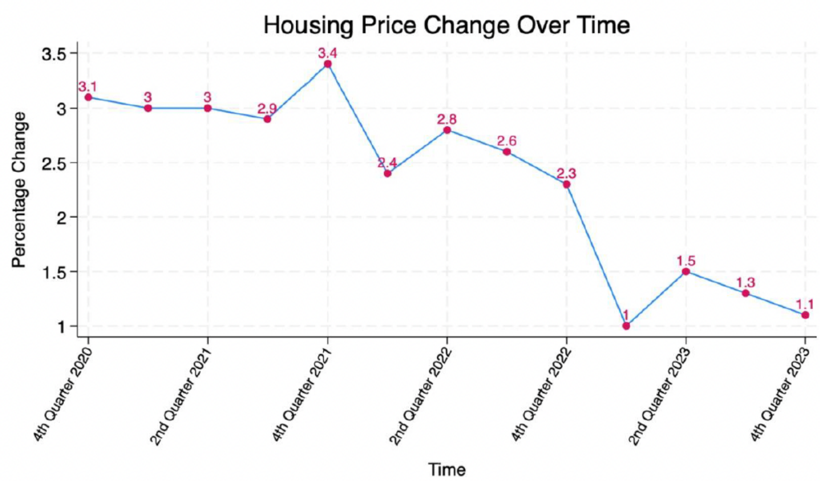
Over the past few years, HDB resale flat prices have risen steadily. The Resale Price Index (RPI), which tracks overall price movements in Singapore's public housing resale market, increased from **138.1** in the fourth quarter of 2020 to **180.4** in the fourth quarter of 2023, an overall rise of about **30%**.



Source: hdb.gov.sg; The RPI can be used to compare the overall price movements of HDB resale flats. It is calculated using resale transactions registered across towns, flat types, and models. The base period is the 1st quarter of 2009, i.e. RPI has a value of 100 in 1st Quarter 2009. For example, if the index increases from 100 to 110 in 1 year, that means that overall, HDB resale flat prices increased by about 10% over that year.

(b) T2: Cooling Housing Price Growth

Since 2020, the pace of resale flat price growth has slowed markedly. After peaking at **3.4%** in the fourth quarter of 2021, the growth rate fell to just **1.1%** by the fourth quarter of 2023, indicating a clear cooling of the market.



Source: hdb.gov.sg

Notes: This figure presents the information provided to respondents in the housing price treatments. Panel (a) displays the level of the HDB Resale Price Index, highlighting the cumulative increase in public housing resale prices since 2020. Panel (b) displays the quarterly growth rate of the same index, emphasizing the deceleration in price growth in recent years. All figures are based on official statistics from the Housing & Development Board in Singapore.

Figure 2: Housing Policy Information Treatments

(a) T3: Subsidy Treatment

The Enhanced CPF Housing Grant (EHG) is a government subsidy in Singapore that helps first-time home buyers afford an HDB flat. During the 2024 National Day Rally, it was announced that the Enhanced CPF Housing Grant (EHG) will be increased for eligible families purchasing HDB flats.

	Current	**Revised**
Enhanced CPF Housing Grant	Up to $80,000	Up to $120,000

With this change, the maximum total grant available to first-time families buying resale flats will rise to **$230,000**, comprising the EHG of up to $120,000, the CPF Housing Grant of up to $80,000, and the Proximity Housing Grant of up to $30,000.

Source: hdb.gov.sg

(b) T4: Tax Treatment

Property tax for owner-occupied residential properties applies to condominiums, HDB flats, and other homes where the owner lives in ("occupies") the property. From January 2024, property tax rates have been revised upwards for most residential properties.

	Current	**Revised**
Property Tax Rates	Up to 23%	Up to 32%

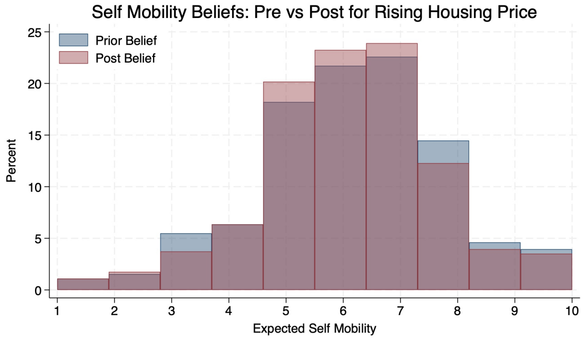
For example, for an owner-occupied property with an annual value (AV) of $100,000, the **property tax payable will increase from $8,730 to $11,980.**

*The AV of buildings is the estimated gross annual rent of the property if it were to be rented out, excluding furniture, furnishings and maintenance fees.

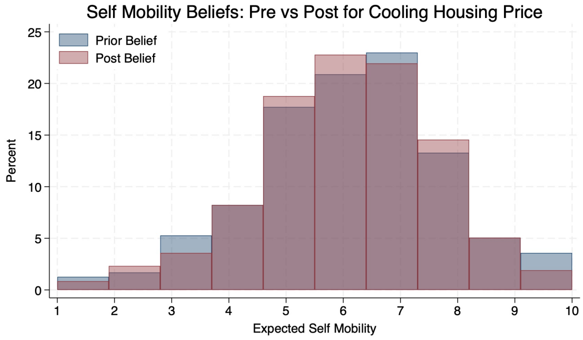
Source: iras.gov.sg

Notes: This figure presents the information shown to respondents in the housing policy treatments. Panel (a) displays information on the Enhanced CPF Housing Grant (EHG), highlighting the announced increase in the maximum grant amount for eligible first-time HDB resale flat buyer. The table reports the current and revised grant ceilings, and the accompanying text describes the resulting increase in total housing grants available to eligible households. Panel (b) presents information on property tax changes for owner-occupied residential properties, including the revision of marginal tax rates and an illustrative example of the increase in tax payable for a property with an annual value of S$100,000.

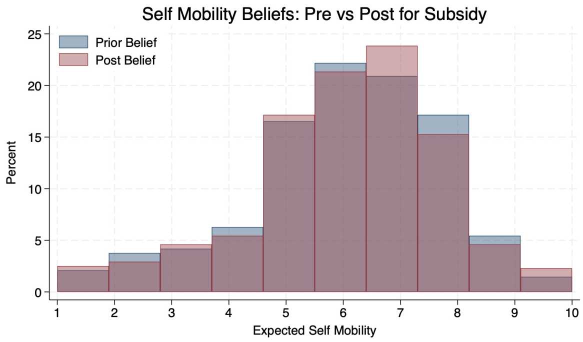
Figure 3: Prior and Posterior Beliefs about Perceived Individual Upward Mobility.



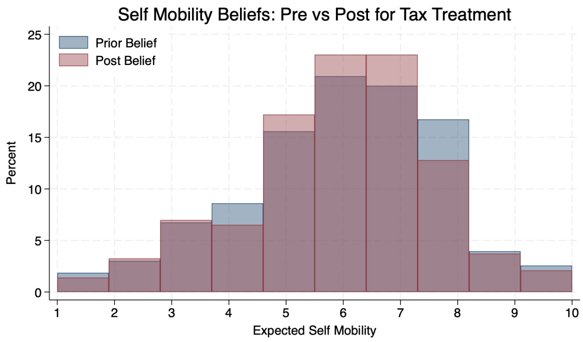
Panel A: Treatment 1



Panel B: Treatment 2



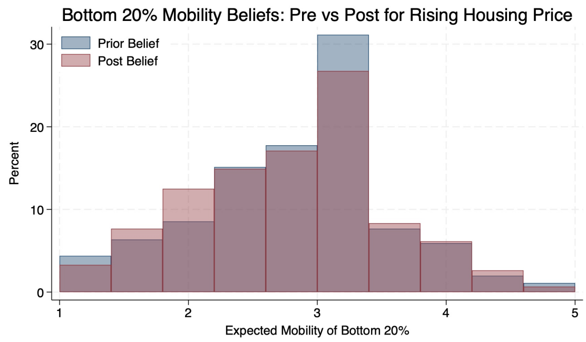
Panel C: Treatment 3



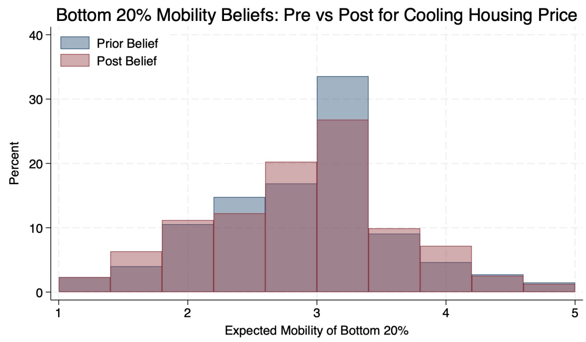
Panel D: Treatment 4

Notes: This figure presents the distribution of respondents' beliefs about their perceived individual upward mobility, measured before and after information exposure. Each panel corresponds to one treatment arm: Panel (a) Rising Housing Price (T1), Panel (b) Cooling Housing Price Growth (T2), Panel (c) Subsidy Treatment (T3), and Panel (d) Tax Treatment (T4). Individual mobility is measured using a ten-rung social ladder, where higher values indicate higher expected future social position. The figure illustrates how different housing market and policy information affects beliefs about respondents' own upward mobility relative to baseline expectations.

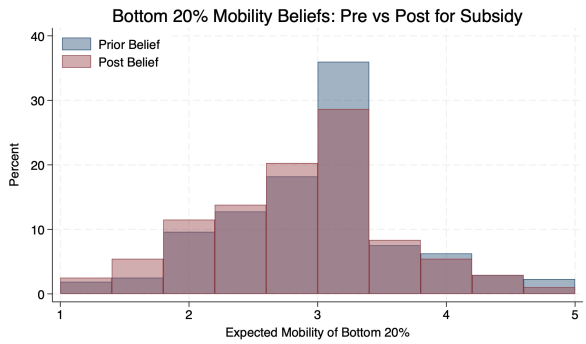
Figure 4: Prior and Posterior Beliefs about Upward Mobility of the Bottom-Origin 20%.



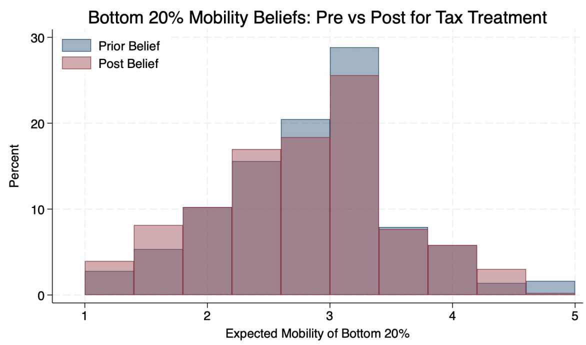
Panel A: Treatment 1



Panel B: Treatment 2



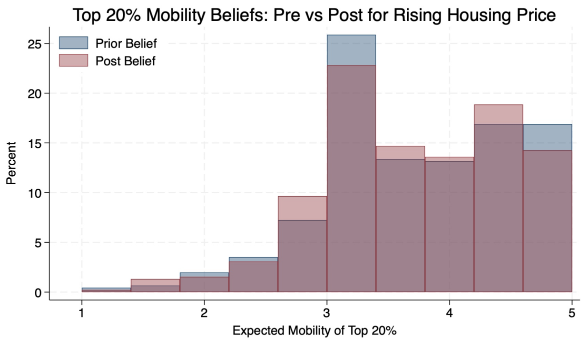
Panel C: Treatment 3



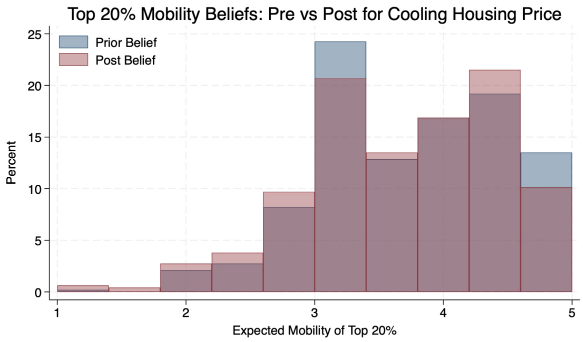
Panel D: Treatment 4

Notes: This figure presents the distribution of respondents' beliefs about intergenerational upward mobility for children born to families in the bottom 20 percent of the social distribution, measured before and after information exposure. Each panel corresponds to one treatment arm, as in the previous figure. Beliefs are elicited using a five-rung social mobility ladder and summarized using an expected-rung measure, which is treated as a continuous variable and grouped into ten equal-width bins for visualization. Higher values indicate greater expected social position.

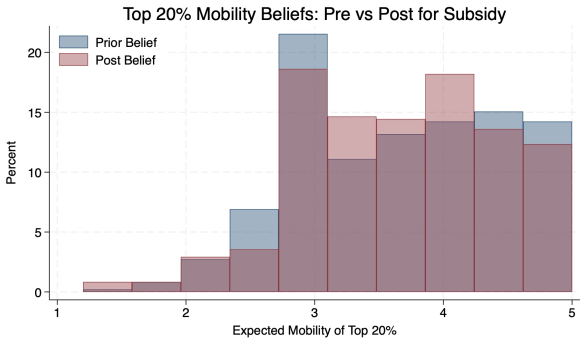
Figure 5: Prior and Posterior Beliefs about Mobility Persistence of the Top-Origin 20%



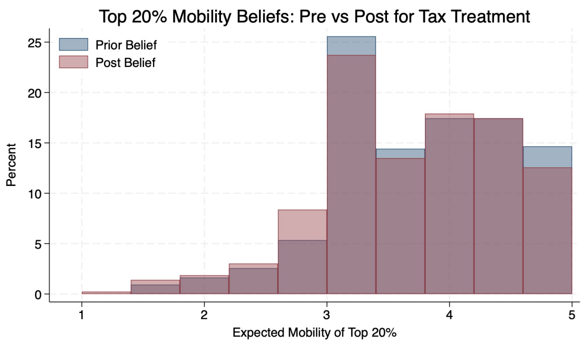
Panel A: Treatment 1



Panel B: Treatment 2



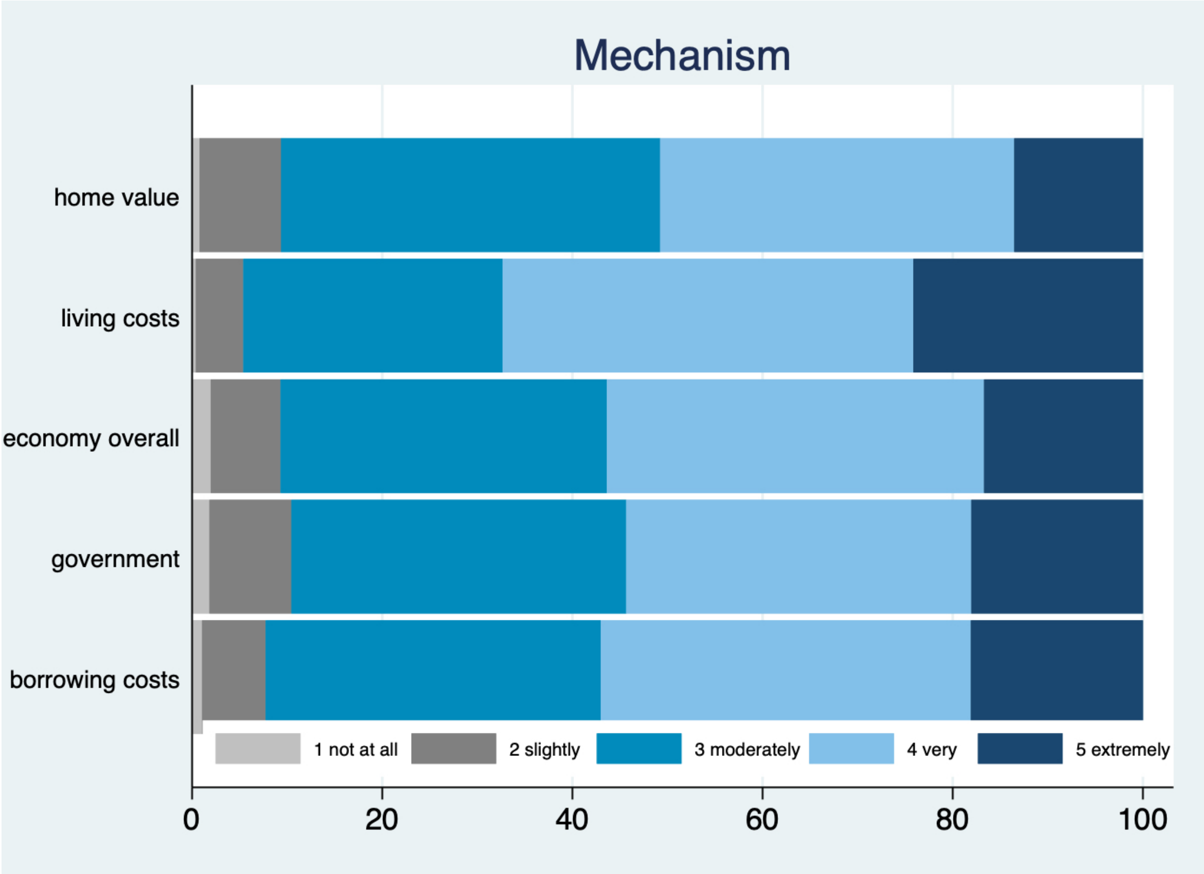
Panel C: Treatment 3



Panel D: Treatment 4

Notes: This figure presents the distribution of respondents' beliefs about intergenerational mobility persistence for children born to families in the top 20 percent of the social distribution, measured before and after information exposure. Each panel corresponds to one treatment arm, as in the previous figure. Beliefs are elicited using a five-rung social mobility ladder and summarized using an expected-rung measure, which is treated as a continuous variable and grouped into ten equal-width bins for visualization. Higher values indicate greater expected social position.

Figure 6: Mechanism



Notes: This figure summarizes respondents' views on how housing conditions affect social mobility through several potential channels, including home values, living costs, borrowing costs, government action, and overall economic conditions. Respondents were asked to assess the extent to which each channel influences social mobility on a five-point scale ranging from 'not at all' to 'extremely.' The bars report the distribution of responses for each channel.

Online Appendix

Housing Markets and the Belief in Opportunity

A Additional Tables and Figures

Table A.1: Test of Balance

	T1		T2		T3		T4	
	Diff	p	Diff	p	Diff	p	Diff	p
Panel A: Demographics								
Age	0.434	(0.601)	1.446*	(0.075)	0.715	(0.392)	0.528	(0.537)
Female	0.027	(0.406)	-0.027	(0.418)	-0.010	(0.769)	0.004	(0.908)
Income	-127.752	(0.754)	554.141	(0.164)	-137.308	(0.738)	283.534	(0.497)
Education	-0.014	(0.577)	0.002	(0.943)	0.011	(0.662)	0.021	(0.405)
Parent Education	-0.048	(0.140)	-0.027	(0.400)	-0.021	(0.504)	-0.012	(0.708)
Employed	-0.018	(0.412)	-0.002	(0.940)	-0.041*	(0.055)	0.001	(0.976)
Liquidity	0.026	(0.305)	0.004	(0.873)	0.011	(0.658)	0.010	(0.704)
Panel B: Prior Expectations								
Bottom Mobility	0.080	(0.105)	0.005	(0.908)	-0.067	(0.157)	0.061	(0.216)
Top Mobility	-0.032	(0.551)	-0.048	(0.345)	0.018	(0.720)	-0.070	(0.173)
Individual Mobility	-0.076	(0.526)	-0.015	(0.900)	0.043	(0.724)	0.134	(0.288)
Housing Price	-0.275	(0.274)	-0.255	(0.311)	-0.369	(0.131)	-0.289	(0.244)
Economic Growth	0.026	(0.903)	-0.091	(0.668)	-0.197	(0.361)	0.037	(0.865)
Inflation	-0.206	(0.338)	-0.085	(0.699)	-0.394*	(0.069)	-0.080	(0.711)
Trust	-0.024	(0.688)	-0.037	(0.524)	-0.014	(0.809)	0.016	(0.795)
N	917		935		939		891	

Notes: This table reports balance tests for baseline covariates across treatment and control groups. Panel A reports demographic characteristics, while Panel B reports pre-treatment expectations. T1-T4 denote the four information treatments. Female is an indicator equal to one if the respondent is female and zero if male. Income is elicited in categories and recoded as a continuous midpoint measure, with no income coded as 0 and the top-coded category assigned SGD 22,000 as in Table 1. High education equals one if the respondent has at least post-secondary education and zero otherwise. Parental education indicators equal one if the respondent's father has at least a post-secondary education and zero otherwise. Employed equals one if the respondent is currently employed and zero otherwise. Liquidity is an indicator equal to one if the respondent reports being able to cover three months of expenses. Prior expectations are constructed from baseline belief measures. Housing price, economic growth, and inflation expectations are constructed as weighted averages from probability-bin responses. Government trust is measured on a 1–5 scale and used as reported. Mobility beliefs are summarized as expected-rung measures from ladder allocations. Statistical significance is based on two-sided t-tests of equality in means between each treatment group and the control group. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table A.2: Pairwise Correlations

	Individual Mobility	Top Mobility	Bottom Mobility	Housing Price	Economic Growth	Inflation	Trust	Age	Income	Education
Individual Mobility	1.000									
Top Mobility	-0.080***	1.000								
Bottom Mobility	0.168***	-0.100***	1.000							
Housing Price	-0.175***	0.235***	-0.053**	1.000						
Economic Growth	0.174***	0.077***	0.138***	0.268***	1.000					
Inflation	-0.176***	0.234***	-0.064***	0.599***	0.336***	1.000				
Trust	0.297***	-0.088***	0.179***	-0.171***	0.164***	-0.177***	1.000			
Age	-0.210***	0.040*	0.065***	0.045**	-0.010	0.087***	0.007	1.000		
Income	0.251***	0.091***	-0.053**	0.068***	0.055***	0.033	0.014	-0.053**	1.000	
Education	0.143***	0.119***	-0.106***	0.050**	-0.018	-0.007	-0.004	-0.242***	0.290***	1.000

Notes: This table reports pairwise Pearson correlation coefficients among baseline mobility beliefs, macroeconomic expectations, and selected respondent characteristics. Individual mobility refers to respondents' expected own social position ten years in the future. Top Mobility and Bottom Mobility refer to expected-rung measures for children born to families in the top and bottom 20% of the social position distribution, respectively. Housing Price, Economic Growth, and Inflation refer to baseline expectations constructed from probability-bin responses. Trust is measured on a 1-5 scale. Income is the midpoint-coded measure of monthly household income, and Education is an indicator for having at least a polytechnic diploma or above. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A.3: Robustness: Treatment Effects on Intergenerational Social Mobility Expectations Using an Alternative Sample

	Bottom-Origin Mobility				**Top-Origin Mobility**			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
T1	-0.105**				0.060			
	(0.048)				(0.050)			
T2		-0.019				0.020		
		(0.046)				(0.049)		
T3			-0.029				0.000	
			(0.046)				(0.049)	
T4				-0.095**				0.013
				(0.047)				(0.049)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
N	946	962	964	952	946	962	964	952
R^2	0.081	0.095	0.101	0.101	0.046	0.059	0.044	0.044

Notes: This table reports OLS estimates of treatment effects on intergenerational mobility expectations under an alternative sample construction. This sample applies only two restrictions: excluding respondents with inconsistent housing tenure or ownership reports and interviews with very short completion times. Columns (1)–(4) use mobility expectations for bottom-origin children as the dependent variable, while columns (5)–(8) use mobility expectations for top-origin children. Each column corresponds to a separate regression comparing one treatment group to the control group. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; and T4 provides information about property taxes. All specifications include the same set of controls: age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, race, and pre-treatment expectations about economic growth conditions and trust. Robust standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table A.4: Treatment Effects on Perceived Individual Upward Mobility

	Perceived Individual Upward Mobility			
	(1)	(2)	(3)	(4)
T1	-0.027			
	(0.099)			
T2		-0.017		
		(0.099)		
T3			-0.064	
			(0.104)	
T4				-0.125
				(0.102)
Controls	Yes	Yes	Yes	Yes
N	917	935	939	891
R^2	0.251	0.240	0.232	0.303

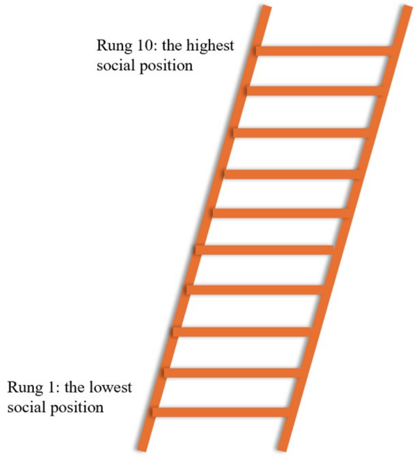
Notes: This table reports OLS estimates of the effects of housing-related information treatments on post-treatment beliefs about participants' individual upward mobility. The dependent variable is respondents' expected own social position ten years in the future. Each column reports results from a separate regression corresponding to a different treatment. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; T4 provides information about property taxes. All specifications include the same set of controls including age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, and pre-treatment expectations about economic growth conditions and trust. Robust standard errors are reported in parentheses. ***, **, and * denote significance at the 1%, 5%, and 10% levels.

Table A.5: Treatment Effects on Perceived Individual Upward Mobility: Heterogeneity by Liquidity

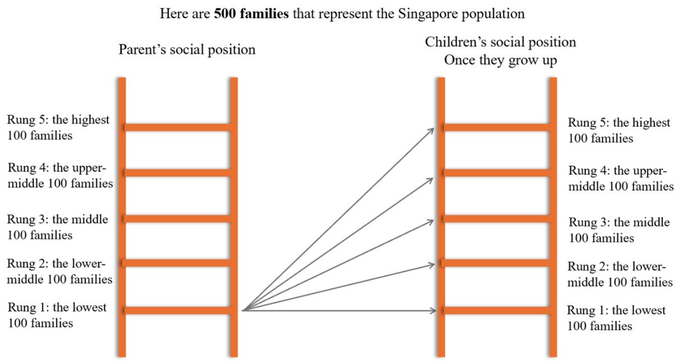
	Perceived Individual Upward Mobility			
	(1)	(2)	(3)	(4)
T1	-0.566***			
	(0.196)			
Liquidity × T1	0.665***			
	(0.199)			
T2		-0.419*		
		(0.224)		
Liquidity × T2		0.477**		
		(0.226)		
T3			-0.847***	
			(0.209)	
Liquidity × T3			0.949***	
			(0.215)	
T4				-0.408*
				(0.228)
Liquidity × T4				0.342
				(0.230)
Controls	Yes	Yes	Yes	Yes
R^2	0.262	0.246	0.252	0.305
N	917	935	939	891

Notes: This table reports OLS estimates of heterogeneous treatment effects on perceived individual upward mobility by liquidity. The dependent variable is respondents' expected own social position ten years in the future. Each column corresponds to a separate regression comparing one treatment group to the control group. T1 provides information about rising housing prices; T2 provides information about cooling housing price growth; T3 provides information about housing subsidies for first-time buyers; and T4 provides information about property taxes. Liquidity is an indicator equal to one if the respondent reports being able to cover three months of expenses. All specifications include the same set of controls, including age, gender, marital status, household size, household income, education, parental education, employment status, housing type, birthplace, race, and pre-treatment expectations about economic growth and government trust. Robust standard errors are reported in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

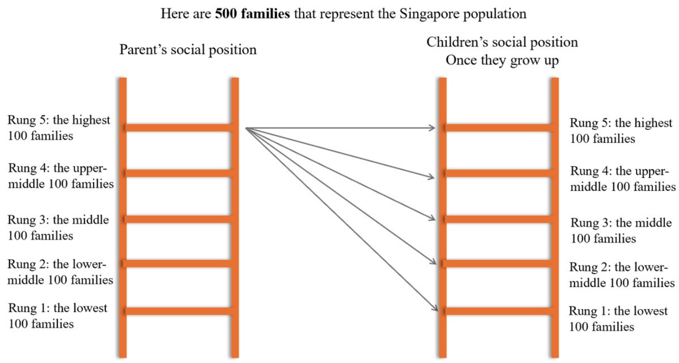
Figure A.1: Social Mobility Ladders



(a) Perceived Individual Upward Mobility Ladder



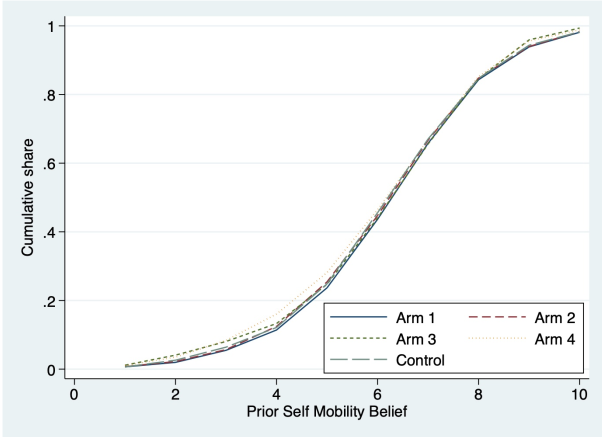
(b) Bottom-Origin Mobility Ladder



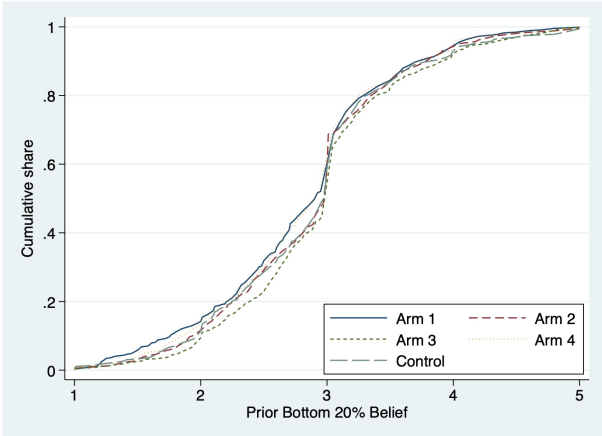
(c) Top-Origin Mobility Ladder

Notes: This figure illustrates the social mobility ladder framework used in the survey. Panel (a) presents the perceived individual upward mobility ladder, where respondents locate their own expected social position on a 10-rung scale. Panels (b) and (c) depict intergenerational mobility ladders for bottom-origin and top-origin families, respectively. In these panels, 500 families representing the Singapore population are evenly divided across five rungs (100 families per rung) based on parents' social position, and arrows indicate potential mobility outcomes for children once they grow up.

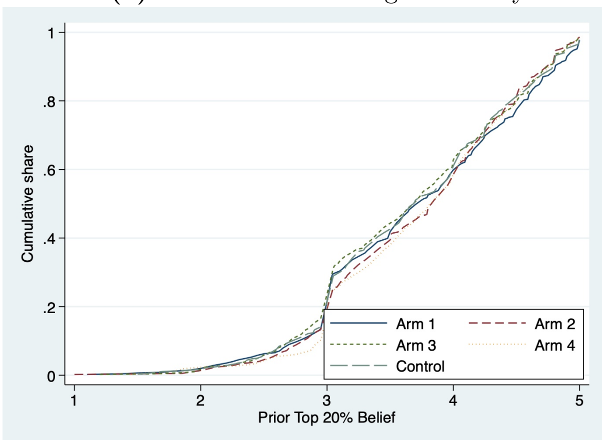
Figure A.2: Test of balance for baseline mobility beliefs



Panel A(1): CDF - Perceived Individual Upward Mobility Priors



Panel A(2): CDF - Bottom-Origin Mobility Priors



Panel A(3): CDF- Top-Origin Mobility Priors

Notes: This figure reports empirical cumulative distribution functions of baseline social mobility beliefs across treatment arms and the control group. Panel A(1) plots the distribution of prior individual upward mobility beliefs, Panel A(2) plots the distribution of prior beliefs about upward mobility for bottom-origin children, and Panel A(3) plots the distribution of prior beliefs about mobility for top-origin children. Each panel compares distributions across all treatment arms and the control group.

Figure A.3: Figure A3 (1)-(4): Test of Balance - Baseline Housing & Macro Expectations

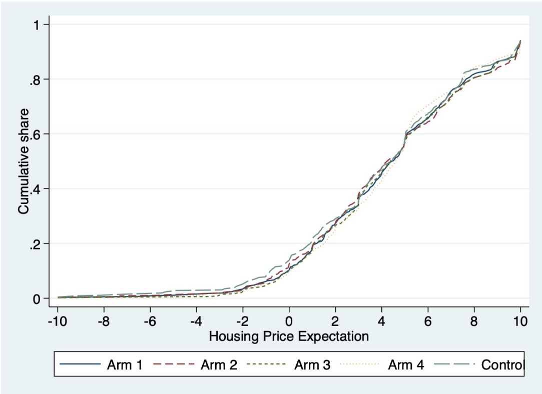


Figure A3(1): CDF — Housing Price Expectations

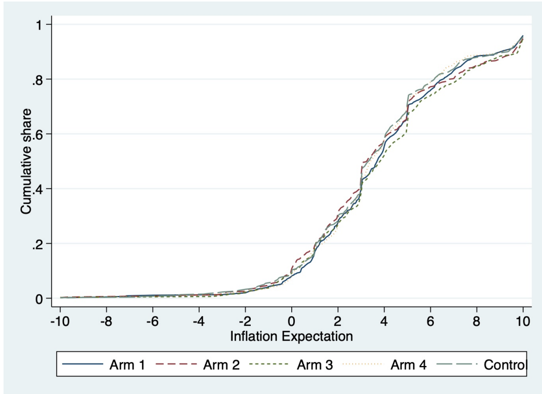


Figure A3(2): CDF — Inflation Expectations

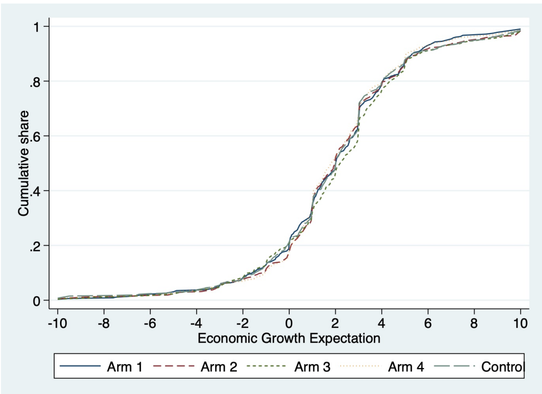


Figure A3(3): CDF — GDP Growth Expectations

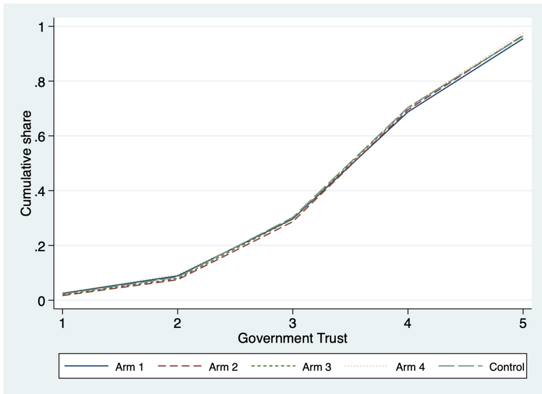
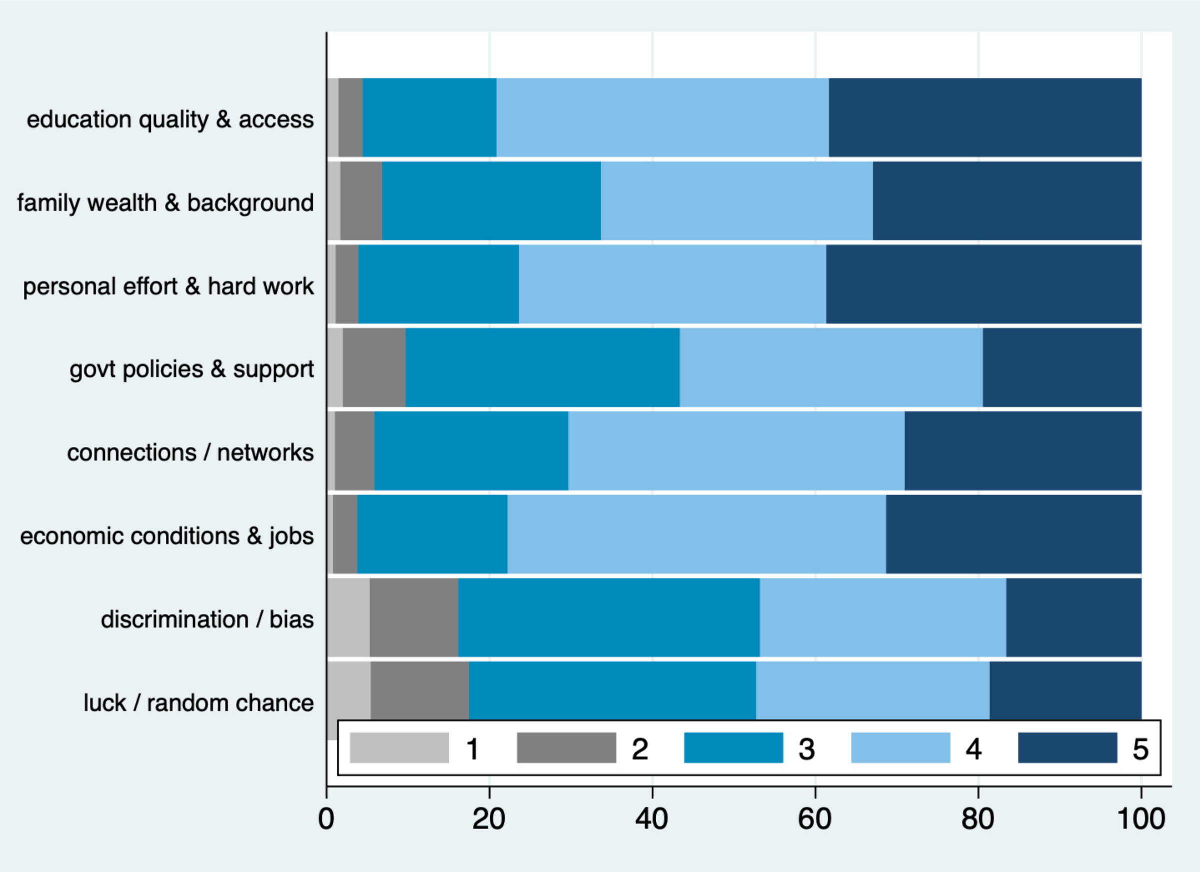


Figure A3(4): CDF — Trust Expectations

Notes: This figure reports empirical cumulative distribution functions of baseline macroeconomic expectations across treatment arms and the control group. Panel A(1) plots prior housing price expectations, Panel A(2) plots prior inflation expectations, Panel A(3) plots prior expectations of economic growth, and Panel A(4) plots baseline government trust expectations. Each panel compares distributions across all treatment arms and the control group.

Figure A.4: Factors Influencing Social Mobility



Notes: This figure reports respondents' baseline perceptions of factors influencing social mobility prior to treatment exposure. Respondents rated the importance of each factor on a five-point scale ranging from 1 (not at all important) to 5 (extremely important). Bars show the distribution of responses for each factor.

B Proof

B.1 Derivation of (16).

With the linear specification $m(x) = x$ and the location-shift structure $X_{i\delta} = X_{i0} + \delta$, respondent i's reference-dependent utility can be written as

$$U_i(\delta) = \mathbb{E}\big[X_{i0} + \delta + \mu(X_{i0} + \delta - R_{i0})\big] = \mathbb{E}[X_{i0}] + \delta + \mathbb{E}\big[\mu(Z_i + \delta)\big], \tag{1}$$

where $R_{i0} \sim G_{i0}$ and $Z_i \equiv X_{i0} - R_{i0}$ denotes respondent i's baseline comparison gap.

Differentiating with respect to δ yields

$$\frac{dU_i(\delta)}{d\delta} = 1 + \frac{d}{d\delta}\,\mathbb{E}\big[\mu(Z_i + \delta)\big]. \tag{2}$$

Writing the expectation as an integral with respect to the density $h_i(\cdot)$ of Z_i,

$$\mathbb{E}\big[\mu(Z_i + \delta)\big] = \int_{-\infty}^{\infty} \mu(z + \delta)\, h_i(z)\, dz, \tag{3}$$

and using dominated convergence (recall that $\mu(\cdot)$ is piecewise linear and h_i is integrable), we may interchange differentiation and integration to obtain

$$\frac{d}{d\delta}\mathbb{E}\big[\mu(Z_i + \delta)\big] = \int_{-\infty}^{\infty} \mu'(z + \delta)\, h_i(z)\, dz = \mathbb{E}\big[\mu'(Z_i + \delta)\big]. \tag{4}$$

From the gain–loss function

$$\mu(y) = \begin{cases} \eta y, & y \geq 0, \\ \eta\lambda y, & y < 0, \end{cases} \qquad \eta > 0,\ \lambda > 1,$$

the derivative satisfies

$$\mu'(y) = \begin{cases} \eta, & y > 0, \\ \eta\lambda, & y < 0. \end{cases}$$

Since the distribution of Z_i is continuous, $\Pr(Z_i + \delta = 0) = 0$, and the nondifferentiability at zero does not affect the expectation.

Therefore,

$$\begin{aligned} \mathbb{E}\big[\mu'(Z_i + \delta)\big] &= \eta\ \Pr(Z_i + \delta \geq 0) + \eta\lambda\ \Pr(Z_i + \delta < 0) \\ &= \eta\big(1 - H_i(-\delta)\big) + \eta\lambda\, H_i(-\delta) \end{aligned} \tag{5}$$

where $H_i(\cdot)$ denotes the cumulative distribution function of Z_i.

Substituting (5) into (2) yields

$$\frac{dU_i(\delta)}{d\delta} = 1 + \eta \; \Pr(Z_i + \delta \geq 0) + \eta\lambda \; \Pr(Z_i + \delta < 0) = 1 + \eta + \eta(\lambda - 1)H_i(-\delta), \quad (6)$$

which establishes (16).

B.2 Derivation of (17) and (18)

From (16), we have

$$\frac{dU_i(\delta)}{d\delta} = 1 + \eta\Big(1 + (\lambda - 1)H_i(-\delta)\Big), \quad (7)$$

where $H_i(\cdot)$ is the CDF of respondent i's comparison gap $Z_i = X_{i0} - R_{i0}$. Differentiating (7) once more with respect to δ yields

$$\frac{d^2U_i(\delta)}{d\delta^2} = \eta(\lambda - 1)\,\frac{d}{d\delta}H_i(-\delta). \quad (8)$$

By the chain rule,

$$\frac{d}{d\delta}H_i(-\delta) = H_i'(-\delta) \cdot \frac{d(-\delta)}{d\delta} = h_i(-\delta) \cdot (-1) = -h_i(-\delta), \quad (9)$$

where $h_i(\cdot) = H_i'(\cdot)$ denotes the density of Z_i (which exists under the maintained regularity conditions). Substituting (9) into (8) gives

$$\frac{d^2U_i(\delta)}{d\delta^2} = -\eta(\lambda - 1)\,h_i(-\delta), \quad (10)$$

which is (17).

Evaluating (10) at the benchmark $\delta = 0$ yields

$$\left.\frac{d^2U_i(\delta)}{d\delta^2}\right|_{\delta=0} = -\eta(\lambda - 1)\,h_i(0). \quad (11)$$

Defining respondent i's fragility as $\varphi_i \equiv h_i(0)$ implies

$$\left.\frac{d^2U_i(\delta)}{d\delta^2}\right|_{\delta=0} = -\eta(\lambda - 1)\,\varphi_i, \quad (12)$$

which establishes (18).